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A comprehensive review on advancements and challenges of deep learning in tunnel and underground engineering

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Abstract

This review explores the integration of deep learning in tunnel and underground engineering, focusing on areas like parameter inversion, machinery optimization, risk assessment, and safety monitoring. The study highlights how deep learning has advanced tunnel parameters' prediction and optimization, improving construction machinery's accuracy and efficiency. Additionally, integrating multi-source data has enabled innovative risk control solutions and dynamic construction management. Advanced image processing and computer vision, combined with deep learning, have enhanced real-time defect detection, significantly improving safety and maintenance. Despite advancements, challenges remain in deep learning applications in tunnel engineering, particularly in data sharing, the absence of standardized metrics, and real-time data processing in complex geological conditions. Looking forward, future research should focus on refining deep learning algorithms for more accurate and robust predictions, improving system integration, and developing intelligent disaster prevention systems for tunnel. Ultimately, deep learning is poised to be a cornerstone in the evolution of smart, sustainable, and resilient tunnel construction practices, aligning with global objectives for intelligent infrastructure development.

Keywords Tunnel and underground engineering, Deep learning, Multivariate monitoring data, Lining disease identification, Intelligent disaster prevention system

1 Introduction

Since the early twenty-first century, tunnel and underground engineering has increasingly focused on overcoming complex geological conditions, particularly in mountainous regions, to improve transportation networks [1, 2]. Notable examples include the Gotthard Base Tunnel in Switzerland-the world's longest and deepestand extensive systems like Japan's Seikan Tunnel and South Korea's tunnel [3–5]. In China, the emphasis has shifted from urban plains to mountainous areas, supporting regional economic growth and expanding national transportation infrastructure. By the end of 2023, China

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had built 27,297 highway tunnel spanning 30,232 km, including 25 extra-long tunnel over 10 km [6–8], and 18,573 railway tunnel totaling 23,508 km [9]. Additionally, Urban rail transit systems are operational in 59 cities, with 11,123.65 km of track, reflecting rapid expansion, particularly in central and western regions [10].

However, tunnel design and construction are challenging because geological and hydrogeological conditions are highly variable, heterogeneous, and uncertain [11]. These environments, along with interactions between the excavated mass, surrounding geological conditions, and machinery, can lead to hazardous events like rock bursts, water inrushes, and surface subsidence [12, 13]. Understanding the mechanical behavior, stability, and interactions between geotechnical media and engineering structures is critical [14]. Factors influencing tunnel construction, such as geological conditions, hydrogeological parameters, and in-situ stress states, span multiple scales and levels [15, 16]. While numerical methods like finite element and discrete element modelling are commonly used to address these challenges [17, 18], data acquisition technologies still face constraints, making it difficult to obtain accurate tunnel face feature data during excavation [19]. Moreover, a significant portion of collected data remains underutilized due to the lack of effective analysis tools [20, 21]. Data mining techniques, such as data augmentation and complex dataset recognition, have been successfully applied to mitigate geological hazards [22–24], advance geological forecasting, and predict surrounding rock deformation [25]. Advancements in computer technologies and Artificial Intelligence (AI) have led to the widespread use of Machine Learning (ML) in engineering, particularly in feature extraction and pattern recognition [26, 27]. Traditional ML methods struggle with large, complex datasets due to their reliance on manual feature engineering [28, 29]. In contrast, Deep Learning (DL) automates feature extraction using multi-layer neural networks, improving both efficiency and accuracy [30, 31]. This makes DL particularly well-suited for tunnel engineering, where it addresses challenges like nonlinearity and complexity. DL has been applied to areas such as parameter inversion [32–36], disaster prediction [37–39], and risk assessment [40–42], enhancing traditional models and supporting geological exploration, construction monitoring, and long-term maintenance. As shown in Fig. 1, in tunnel and underground engineering, DL enhances traditional modelling approaches and supports geological exploration, construction

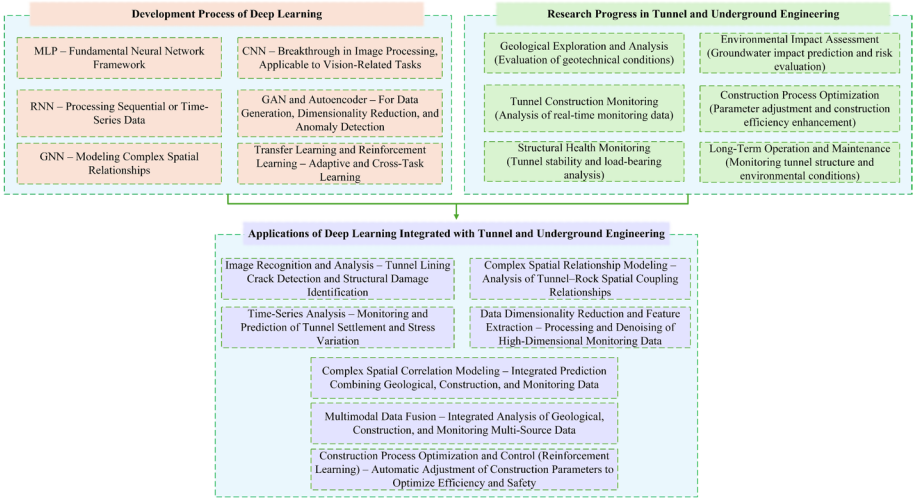


Fig. 1 Deep learning applications in tunnel and underground engineering [20]

monitoring, and long-term maintenance. The integration between DL and underground engineering has become increasingly comprehensive. From the foundational Multilayer Perceptron (MLP) and Convolutional Neural Network (CNN) to advanced models such as Graph Neural Networks (GNNs), Generative Adversarial Networks (GANs), and Reinforcement Learning (RL), DL architectures provide powerful tools for automated perception, intelligent prediction, and adaptive optimization. When applied to various stages of tunnel engineering, these algorithms enable data-driven decision-making through feature extraction, time-series analysis, and multimodal information fusion, thereby improving the efficiency, safety, and sustainability of underground construction and operation [43, 44].

Additionally, DL supports the preprocessing of simulation data and parameter optimization and improves the accuracy of traditional analytical models. This integration not only boosts the performance of numerical simulations but also contributes to the safety and efficiency of tunnel engineering projects [20].

At present, research on the application of DL algorithms in tunnel and underground engineering remains largely confined to specific and isolated subfields. Comprehensive and systematic reviews that outline the overall development and application of DL in this domain are relatively scarce. In response to this gap, this paper presents an overview of the evolution of DL in tunnel and underground engineering, reviews the body of literature-both domestic and international-published since the advent of DL, and categorizes its applications into seven primary research directions: parameter inversion analysis, prediction and optimization of construction machinery parameters, control of construction and operational processes with risk assessment, tunnel safety monitoring and defect detection, structural health prediction of tunnel, surrounding rock classification, and tunnel face image recognition and classification. This paper explores and summarizes the intelligent applications of DL across these seven areas, identifies current research limitations and technical challenges, and provides an outlook on the future development of DL in intelligent, safe, and sustainable tunnel and underground engineering.

While previous reviews such as Liu et al. [45] and Wu et al. [46] have explored the application of AI and ML in tunnel and underground engineering, this review offers several novel contributions. First, it presents a comprehensive overview of the integration of DL techniques across multiple, distinct subdomains of tunnel engineeringranging from parameter inversion and construction machinery optimization to risk assessment, defect detection, and structural health monitoring. Unlike previous reviews, which often focus on a single application or a narrow set of techniques, this work provides a cross-domain comparison, assessing different DL algorithms, dataset sizes, performance metrics, and key findings. Second, we include a thorough analysis of the challenges and limitations that have been under-explored in past reviews, particularly regarding data quality, algorithm reliability, and real-time implementation in complex geological environments. Finally, this review emphasizes future research directions such as the potential for AI integration with Digital Twin (DT) technologies, intelligent disaster prevention, and real-time monitoring, thus presenting a forward-looking perspective on the field. These aspects set this review apart, providing readers with a more holistic understanding of the current state and future potential of DL in tunnel and underground engineering.

Unlike previous AI-related reviews that present descriptive taxonomies, this work establishes a mechanistic synthesis linking DL architectures to the distinct

spatiotemporal, multi-physics, and nonlinear characteristics inherent in tunnel engineering systems. Specifically, the review formulates an algorithm-to-process mapping framework, demonstrating how CNNs exploit spatial dependencies for lining defect detection, recurrent architectures: Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit.

(GRU) capture temporal evolution of excavation dynamics, and GANs enable synthetic augmentation of geotechnical datasets with sparse field measurements. Furthermore, by integrating multi-source sensing data (e.g., Ground Penetrating Radar (GPR), acoustic emission, micro-seismic, and Closed-Circuit Television (CCTV) imagery, this review provides a unified conceptual pathway toward "Physics-Aware DL for Subsurface Systems." This synthesis contributes a domain-specific analytical lensinterpreting algorithmic performance through tunnel physics, construction-phase uncertainty, and material heterogeneitythereby distinguishing the present work from general AI reviews in civil infrastructure and establishing its theoretical originality within the tunnel discipline.

2 Deep learning development in tunnel and underground engineering

Early foundational work in this area can be traced to Flood et al., who was among the first to apply neural networks to civil engineering, particularly for optimizing construction workflows [47]. Tunnel construction involves excavating media that are naturally formed and geologically complex, making the process inherently dynamic in both time and space [48]. Therefore, simulating and analyzing the stress conditions of the surrounding rock and tunnel lining during excavation is essential for the design of dynamic support systems, which in turn enables the prediction of structural stability [49, 50]. However, due to the intricate factors affecting the stability of tunnel structures during both construction and operational phases, current assessments still heavily rely on expert knowledge and engineering experience. With the advancement of computing technologies, AI-driven intelligent analysis has become increasingly feasible. Liu et al. reviewed the emerging trends and research frontiers of AI in tunnel construction, emphasizing its critical role in the development of intelligent tunnel systems, risk management, and automated behavior monitoring [45].

A comprehensive literature review was conducted using academic databases, primarily the Web of Science Core Collection, covering the past ten years. The search employed the keywords: "tunnel engineering" OR "underground space" AND "deep learning". After excluding studies with low relevance, a total of 494 peer-reviewed English-language publications were included for analysis. Figure 2 illustrates the systematic screening process, aligned with PRISMA guidelines. An initial 1,617 articles were retrieved. After filtering for relevance and publication year, 1,220 records remained. The subsequent removal of non-English and non-journal/conference entries reduced the count to 1,150. A two-stage eligibility assessment followed: 336 articles were excluded based on title and abstract screening, and an additional 320 were removed after full-text review due to insufficient relevance or methodological limitations. In addition to relevance-based screening, methodological robustness and validation characteristics were considered during full-text assessment. Studies were evaluated with respect to the clarity of dataset description, transparency of model configuration and evaluation metrics, and the type of validation adopted (simulation-only, hybrid numerical–experimental verification, or field-based validation using real project datasets). This distinction enabled a structured

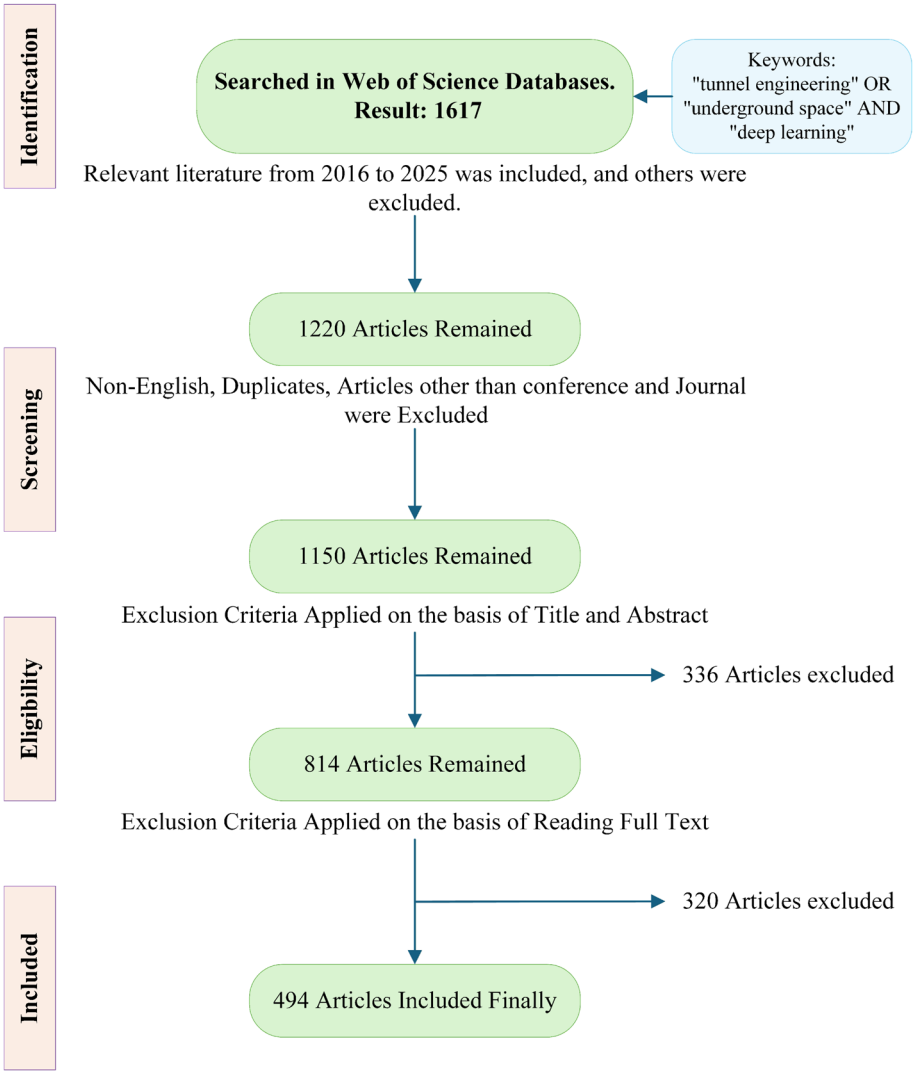


Fig. 2 Article screening and selection process

interpretation of evidence maturity and practical applicability within tunnel engineering contexts. The final dataset of 494 articles formed the evidence base for this review, with the validation-level differentiation informing the critical synthesis and discussion presented in Sects. 4.1 and 4.2.

Figure 3 shows a steady upward trend in publications since the introduction of DL in 2006, with a notable surge after 2019. This increase can be attributed to growing recognition of deep learning's potential, advancements in AI and computational tools, more international collaborations, and increased research funding for large-scale infrastructure projects. Each publication was classified according to the Scimago Journal Rank (Q1–Q4 and unranked) to reflect the distribution of research quality. Publication output rose from 9 papers in 2016 to a peak of 105 in 2024, with 99 publications recorded up to November 2025, indicating that the upward trend remains ongoing. The marked post-2019 surge also reflects the widespread adoption of GPU-accelerated computing, the availability of open-source AI frameworks, and the proliferation of sensor-driven tunnel monitoring and digital-twin systems that enabled large-scale data generation. The increasing share of Q1–Q2 publications since 2021 demonstrates a transition from

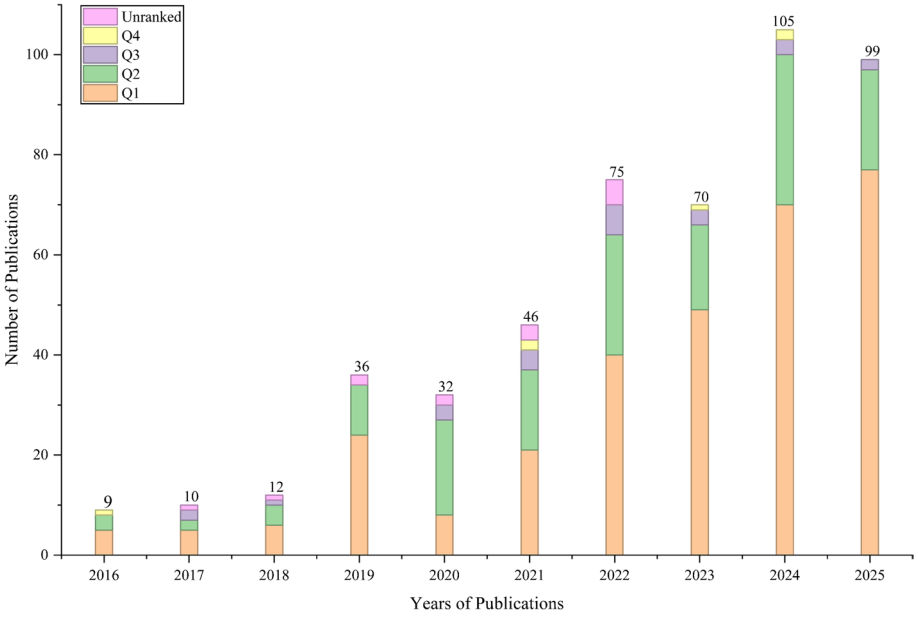


Fig. 3 Distribution of published years of relevant literature

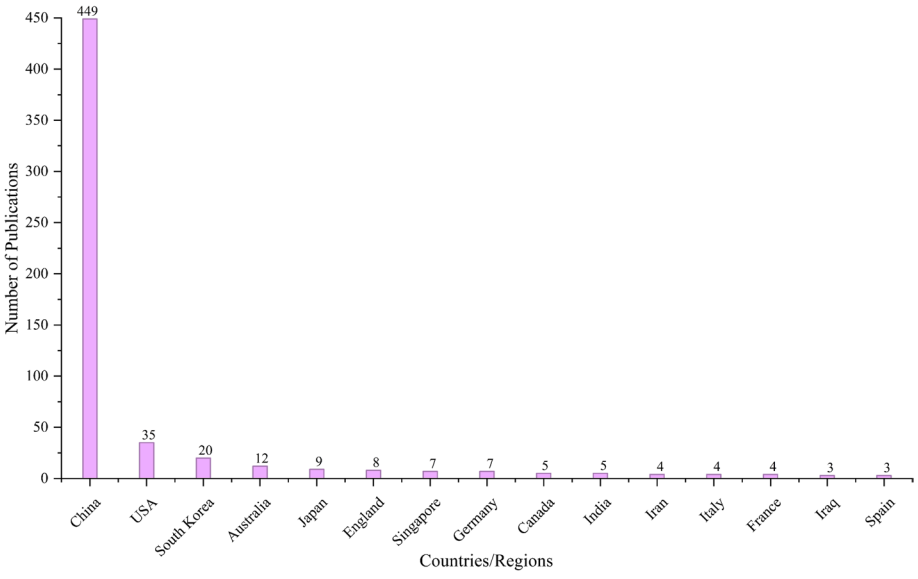


Fig. 4 The geographic distribution of published articles

exploratory studies to rigorously validated, application-oriented research, signalling the growing maturity of DL applications in tunnel and underground engineering.

Based on the analysis of the top 15 contributing countries retrieved from the Web of Science databases, the geographic distribution of research articles highlights significant international engagement in the application of DL for tunnel and underground engineering. As depicted in Fig. 4, China overwhelmingly leads with 449 publications, accounting for the vast majority of the literature, indicating its dominant role and intensive research activity in this domain. The United States follows with 35 publications, reflecting substantial, though comparatively lower, contributions. Other key countries include South Korea (20 articles), Australia (12), Japan (9), England (8), Singapore and Germany (7

each), Canada and India (5 each), Iran, Italy, and France (4 each), Spain and Iraq (3 each). This distribution underscores the global relevance of the topic, with a strong concentration of research in East Asia, North America, and select European nations. It also reflects national priorities in advancing intelligent construction technologies and addressing the complex challenges posed by subsurface engineering environments.

The integration of DL in tunnel engineering has progressed alongside significant advancements in neural network architectures. To evaluate the current state of DL applications in this field, the following Table 1 summarizes and compares the algorithms, datasets, performance metrics, findings, and limitations of the studies in each application domain.

The table presents representative algorithms, data characteristics, performance indicators, and limitations across key application domains, highlighting comparative insights and research gaps in predictive modelling, monitoring, and optimization. Although DL techniques have achieved remarkable progress across multiple domains in tunnel and underground engineering, persistent challenges constrain their broader applicability and generalization. In parameter inversion analysis, advanced models such as ES-Caps-FCN and BP–FLAC3D hybrids have demonstrated exceptional predictive performance ($R^2 > 0.99$) and computational efficiency, being up to 200 times faster than traditional numerical simulations [51, 52]. However, these frameworks remain limited by the reliance on synthetic or simulation-generated datasets, restricted feature diversity, and a lack of validation using field-scale experimental data. Similarly, in construction machinery optimization, time-series models such as LSTM–GWO and 1D-CNN architectures have achieved high accuracy ($R^2 \approx 0.97–0.98$) in predicting tunnel parameters, yet their generalization is constrained by project-specific data sparsity and limited geological variability [53, 54]. In defect detection, deep convolutional architectures such as YOLOv5, YOLOv8, and knowledge-distilled teacher–student networks have reached detection accuracies exceeding 85–95% while significantly reducing model size and improving small-target identification [57, 58]. Nonetheless, issues such as domain shift, limited image diversity, and performance degradation under varying illumination or occlusion conditions continue to hinder model robustness in real-world inspection scenarios. These findings collectively highlight the need for standardized, multi-site datasets, cross-domain validation frameworks, and adaptive model architectures capable of maintaining interpretability and performance consistency under heterogeneous operational environments. The MLP, introduced in the 1980s, laid the groundwork for deep neural networks, enabling the development of DL algorithms [65]. A major breakthrough came in 2006 with Hinton's Deep Belief Network (DBN), addressing key optimization issues in deep architectures [66]. This advancement transformed neural networks from simple models into complex, multi-task systems, sparking global interest in DL [67]. Since then, various models have emerged, including CNNs like Residual Network (ResNet), GoogLeNet, and YOLO; RNNs such as LSTM and GRU; GANs; autoencoders; GNNs; and frameworks for transfer and RL [68–70], marking the rise of intelligent computing in engineering. While DL architectures such as CNNs, RNNs, and GNNs have been extensively applied across diverse domains, their adaptation to tunnel and underground engineering demands rigorous domain-specific reformulation. The inherently nonlinear, heterogeneous, and spatiotemporal characteristics of geotechnical data necessitate algorithmic modifications beyond conventional implementations. For example, CNN-based

models for tunnel defect detection frequently employ dilated convolutions and residual connections to expand receptive fields and enhance spatial feature extraction in limited and noisy image datasets [71]. Similarly, RNN variants such as Bi-directional Long Short-Term Memory (Bi-LSTM) and GRU are optimized to capture temporal dependencies in excavation dynamics through adaptive learning rate scheduling and dropout regularization strategies [72]. Mathematically, DL models can be generalized as nonlinear function approximators, as expressed in Eq. (1):

$$f_\theta(x) = y \tag{1}$$

In Eq. 1, x denotes the input vector and θ represents the trainable parameters via Stochastic Gradient Descent (SGD) and backpropagation. This formulation enables the model f_θ to learn complex, nonlinear mappings between input data and target outputs, serving as the foundation for various predictive and inverse modelling tasks in tunnel and underground engineering.

To optimize the model, a loss function $\mathcal{L}(\theta)$ is defined to quantify the discrepancy between the predicted output y_i and the true target y, as shown in Eq. (2):

$$L(\theta) = \frac{1}{N} + \sum_{i=1}^{N} \ell\left(y_i - f_\theta(x_i)\right)^2 \tag{2}$$

In Eq. 2 N denotes the number of training samples, and ℓ represents a specific loss metric such as the Mean Squared Error (MSE) or cross-entropy loss, depending on the task. The optimization process iteratively updates the parameters θ in the direction of the negative gradient of the loss function with respect to θ, as Eq. (3):

$$\theta_{t+1} = \theta_t - \eta 1 B \sum_{i=1}^{B} \nabla_\theta L\left(\theta_t; x_i; y_i\right) \tag{3}$$

In Eq. 3 η is the learning rate controlling the step size during parameter updates. Through repeated forward and backward propagation, the model progressively minimizes $\mathcal{L}(\Theta_t)$, thereby improving its predictive accuracy and generalization capability [73]. Recent tunnel-related studies have also incorporated multi-objective optimization frameworks for hyperparameter tuningbalancing prediction accuracy, computational efficiency, and physical interpretabilitythrough metaheuristic algorithms such as Particle Swarm Optimization (PSO), Bayesian optimization, and genetic algorithms [74, 75].

Beyond deterministic modelling, Uncertainty Quantification (UQ) is indispensable in DL -driven geotechnical analysis. Techniques including Monte Carlo dropout, ensemble learning, and Bayesian neural networks (BNNs) have been adopted to characterize both epistemic uncertainties, arising from limited or sparse geological samples, and aleatoric uncertainty, induced by measurement noise and environmental variability. Dataset quality and representativeness further require systematic evaluation: tunnel engineering datasets are often affected by severe class imbalance (e.g., rare defect instances), spatial heterogeneity across excavation segments, and temporal drift associated with evolving geological or operational conditions. Mitigating these limitations involves advanced data augmentation via GANs or Synthetic Minority Oversampling (SMOTE), coupled with stratified k-fold cross-validation and transfer learning across geological domains to enhance generalizability and robustness [76–78]. Collectively, these methodological

Table 1 Summary of deep learning applications in tunnel and underground engineering (2016–2025)

Application domain	Algorithms used	Data size/Scale	Performance metrics	Key findings	Limitations	Refs.
Parameter inversion analysis	ES-Caps-FCN, DNN, LR	877 cases (~10.78 M samples)	MSE, MAE, R^2, MRE	ES-Caps-FCN achieved lowest MSE = 0.059866%, MAE = 1.806252%, R^2 = 99.813%, outperforming DNN and LR; DL model 200 × faster than FLAC3D (1 min vs. 3 h)	Limited features used; trained on simulated data only; lacks physical experiment validation; high training computation cost	[51]
	BP Neural Network with FLAC3D simulation	600 samples	RE, MSE, Correlation Coefficient	BP–FLAC3D inversion yields high accuracy (R^2 = 0.991); relative error of stress ≤ 5%; faster convergence vs. traditional iterative methods	Limited to single geological model; relies on synthetic (not field) data; generalization to complex geology unverified	[52]
Construction machinery optimization	LSTM–GWO; compared with LSTM, KNN, GPR, SVR, DT	1,125 samples	RMSE, MAPE, R^2	LSTM–GWO achieved R^2 = 0.9795, RMSE = 0.004 m/h, MAPE = 0.009%; outperformed LSTM, SVR, GPR, KNN, DT; robust against time-series length; training time 32.6 s	Single-project data; limited dataset (1125 samples); relies on numerical and lab data only; generalization to other geology untested	[53]
	1D CNN; compared with MLP, LSTM, ExtraTree, LightGBM, GBM, RF, AdaBoost, k-NN	726 samples	R^2, MSE, MAE	1D CNN achieved R^2 = 89.6%, MSE = 57.5, MAE = 1.6; computation time 3 min 22 s; best among all compared models	Based on single project (Japan); limited to soft ground; lacks generalization across geology	[54]
Risk assessment & safety monitoring	CNN (Xception) + Bayesian Inference	34,372 images	Accuracy, precision, recall	Bayesian–CNN hybrid achieved 81.3% accuracy, 100% recall, 94% precision; enhanced interpretability and reliability vs CNN-only	Limited to NATM tunnel; dataset small; weak performance in some classes; generalization to other tunnel methods unverified	[55]
	SVR, ANN, Hybrid SVR–ANN	9 input parameters (real-time TBM data)	R^2, RMSE, VAF	Hybrid SVR–ANN achieved R^2 = 0.97, RMSE = 0.011 for advance rate prediction; effective integration of ML and DT for real-time monitoring and visualization	Single case study; limited geological diversity; lacks validation on multiple TBM types; real-time data fusion complexity not fully addressed	[56]
Defect Detection	Improved YOLOv5 with Knowledge Distillation, C3CSFM, MDFPN, RWNMS	9,422 GPR images	mAP@0.5, model size (MB), FPS	Accuracy improved from 83.4% → 86.5%, model compressed by 80% (16.03 → 3.20 MB), improved small-target detection and real-time performance	Limited dataset and geological diversity; single radar source (TGRI-GPR); slight FPS drop; needs larger dataset and adaptive temperature tuning in distillation	[57]
	YOLOv8 for detection with EfficientNet for classification integrated with BIM	4,500 images	Precision, Recall, F1-score, mAP@0.5	YOLOv8 achieved mAP@0.5 = 95.3%, Precision = 94.7%, Recall = 93.9%; real-time defect detection and BIM-linked visualization improved maintenance efficiency	Dataset limited to one tunnel type; real-world lighting variations affect performance; no validation under wet/occluded conditions; BIM integration requires manual tagging	[58]
Tunnel structural health prediction	PIOELM (ELM with CASSA and Pasternak model)	19,000 monitoring points	MSE, RMSE, MAPE	PIOELM reduced MSE by 50.57–85.29%, RMSE by 30.1–61.81%, and MAPE by 18.98–78.54% vs. baselines; achieved strong accuracy, stability, and generalization for complex geotechnical prediction	Pasternak model assumes linear elasticity — limited in capturing nonlinear or plastic soil deformation; lacks long-term prediction capability; future work needed for elastoplastic and time-series modelling	[59]

Table 1 (continued)

Application domain	Algorithms used	Data size/Scale	Performance metrics	Key findings	Limitations	Refs.
	GC-GRU (Graph Convolution with GRU); compared with GRU, LSTM, XGBoost, LR, RNN	2,456 monitoring records	RMSE, PCC	RMSE = 15.52 με (strain), 0.026 mm (opening); PCC ≈ 0.95; improved learning accuracy by 2.2% vs baselines; effective 45-day prediction; outperformed all compared models	Ignores external environmental/load factors (e.g., temperature, water pressure); assumes linear correlation; future work to integrate more influencing factors and longer prediction periods	[60]
Surrounding Rock classification	XGBoost, BPNN, AdaBoost with random forest for feature selection and SMOTE/ADASYN for balancing	500 samples	Accuracy, Precision, recall, F1-score	XGBoost achieved 87.5% accuracy, Precision = 0.90, Recall = 0.88, F1 = 0.89; effectively linked drilling parameters with rock classification for intelligent tunnel decision-making	Small dataset; lacks environmental and geological diversity; limited to drilling/blasting tunnel; future work to include more parameters and sites	[61]
	BILSTM with window selector and voter modules; compared with MLP and SVM	1,300 samples	Accuracy, Precision, Recall, F1-score, AUC	Accuracy = 90%, Precision = 0.88, Recall = 0.87, F1 = 0.88; outperforming MLP (+ 13%) and SVM (+ 15%); strong generalization and robustness for unbalanced, small datasets	Dataset limited to single tunnel (Yunnan); only three rock classes (III–V); ignores environmental and mechanical interaction factors; further validation on larger datasets needed	[62]
Tunnel Face Image Recognition and Classification	U-CliqueNet (U-Net with CliqueNet and attention); compared with FCN, U-Net, SegNet, MFCD	60,000 sub-images	Mean Pixel Accuracy (MPA), Mean IoU (MIoU), Precision, F1-Score	MPA = 92.25%, MIoU = 86.96%, Precision = 86.32%, F1 = 83.40%; outperformed all baselines in hypothesis testing; relative crack width error ≈ − 11.2% – 18.6%	Limited dataset diversity; images from specific lighting conditions; may not generalize to all tunnel types; computationally intensive model	[63]
	Transfer Learning with Inception-V3, ResNet50, Inception-ResNet-V2, MobileNet-V2	3,227 images	Accuracy, Recall	Inception-ResNet-V2 achieved 99.72% accuracy, 99.33% recall; outperformed all models; demonstrated UAV imagery's ability for subsurface geological recognition	Limited dataset size and geological diversity; misclassification caused by shadows and vegetation occlusion; not tested under varying light or weather conditions	[64]

considerations elevate the technical depth of DL applications in tunnel and underground engineering by embedding mathematical rigor, computational efficiency, and uncertainty-aware reasoning within a unified framework of intelligent subsurface modelling.

Research in tunnel and underground engineering from 2016 to 2025 has primarily focuses on six key areas: geological exploration, tunnel construction monitoring, structural health monitoring, construction process optimization, environmental impact assessment, and long-term maintenance [79–82]. DL algorithms are increasingly applied across these domains, progressing from early data classification to advanced intelligent prediction systems [20, 83]. Notable applications include CNN-based detection of lining cracks and structural damage [84], RNN-based tunnel settlement and stress/strain prediction [85], multimodal data fusion for multi-task geological and construction predictions [86], transfer and RL for construction process optimization [87], GAN-based data denoising [88], and GNN-based modelling of tunnel-surrounding rock interactions [89]. The rise of DL in this field is driven not only by technological advancements but also by the need for green, intelligent, and sustainable development in tunnel engineering [90].

3 Current application status of deep learning in tunnel and underground engineering

To ensure conceptual rigor and systematic organization, this review delineates DL applications in tunnel and underground engineering into seven interrelated domains: parameter inversion and geological characterization, prediction and optimization of construction machinery parameters, monitoring of construction processes and structural health, risk assessment and disaster prediction, integration within DT frameworks, intelligent maintenance and inspection, and lithological identification and geological segmentation. These domains were defined according to three core dimensions: the underlying data modality (e.g., image-based, signal-based, or numerical datasets), the learning objective (prediction, classification, or segmentation), and the system integration level (stand-alone model versus coupled DT framework). Collectively, this hierarchical taxonomy establishes a coherent analytical foundation for the subsequent discussion. Moreover, the domains represent a continuum of technological evolution—from perception-driven monitoring toward fully autonomous, decision-support systems-thereby avoiding arbitrary separation and linking each category to a distinct computational and functional aspect of tunnel engineering.

With the advent of the Fourth Industrial Revolution, which emphasizes full intelligent automation, multi-dimensional integration, and eco-friendly development, the concepts of technological innovation and iterative advancement have gradually permeated various branches of engineering [91, 92]. Within this context, DL has emerged as a central theme in AI research. Its application to tunnel and underground engineering problems has become a prominent and expanding trend in both domestic and international academia. The evolution of AI has progressed through several transformative milestones, from its conceptual foundations in theoretical neurology to the emergence of modern deep and agentic learning paradigms. The earliest formal models of neural computation originated with the McCulloch–Pitts neuron, which established the theoretical link between biological cognition and logical computation [93], followed by Rosenblatt's Perceptron, the first trainable artificial neural network, marking the birth of ML [94, 95]. The formal establishment of AI as a scientific discipline at the 1956 Dartmouth

Conference provided the intellectual foundation for computational intelligence research [96]. Subsequent breakthroughs, including the backpropagation algorithm and the Universal Approximation Theorem , enabled the training of multilayer neural networks, initiating the second golden age of AI and advancing nonlinear function approximation [97–99]. The advent of CNNs, exemplified by AlexNet , revolutionized computer vision through hierarchical feature extraction, while RNNs and their gated variants expanded AI capabilities in sequential and time-series modelling [100, 101]. The introduction of the Transformer architecture and subsequent Large Language Models (LLMs)-including Bidirectional Encoder Representations from Transformers (BERT), Generative Pre-trained Transformer (GPT), and ChatGPT-redefined natural language processing and ushered in the multimodal learning era, integrating text, imagery, and video processing by 2023 [102–104]. The reasoning revolution of 2024–2025, led by advanced models such as OpenAI's 0.1 and DeepSeek-R1, introduced higher-order analytical reasoning and contextual adaptation. Most recently, LLM-based agentic AI systems have emerged, capable of autonomous task execution, multi-agent collaboration, and human-aligned decision-making, marking the onset of an era in which AI functions as an intelligent partner rather than a passive computational tool [105–107]. The principal developmental trajectory of neural networks and DL is summarized in Fig. 5.

Parameter inversion analysis uses field-monitored data, such as displacement, stress, and strain, to study rock mass or construction machinery parameters, based on the rock mass constitutive model [108, 109]. Optimizing construction machinery parameters focuses on performance, operational posture, and energy consumption through intelligent prediction and control strategies [110, 111]. In risk assessment, multisource data is leveraged to enable autonomous decision-making during tunnel construction and operation. At the same time, data-driven methods facilitate dynamic evaluation of structural safety during excavation and operation [112, 113]. Tunnel safety monitoring and defect detection utilize on-site imagery to identify damage, such as lining cracks and leaks [114, 115]. Micro-seismic monitoring enables real-time forecasting of hazards, including rockbursts and fires, contributing to an intelligent disaster prevention system [116, 117]. Su et al. (2024) demonstrated the integration of micro-seismic signal monitoring with DL for real-time rockburst detection and classification, as shown in Fig. 6 [20].

Structural health prediction focuses on analyzing tunnel stability during excavation, including predicting deformation trends, material degradation, and long-term mechanical behavior [118, 119]. Surrounding rock classification employs computational methods to identify rock mass parameters, aiding in the determination of surrounding rock

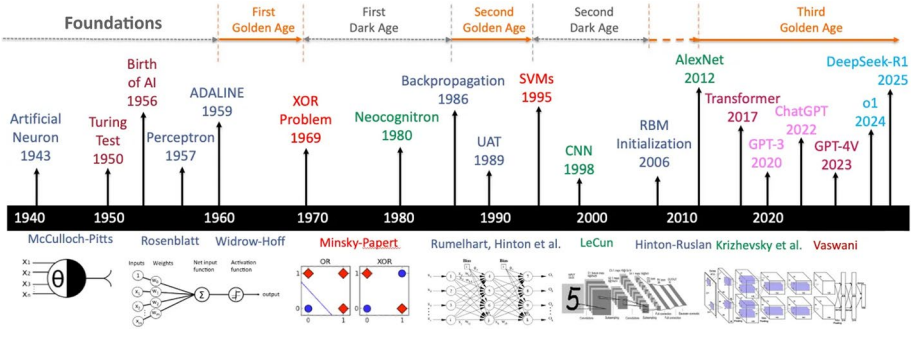


Fig. 5 The development process of DL in tunnel and underground engineering [93–103]

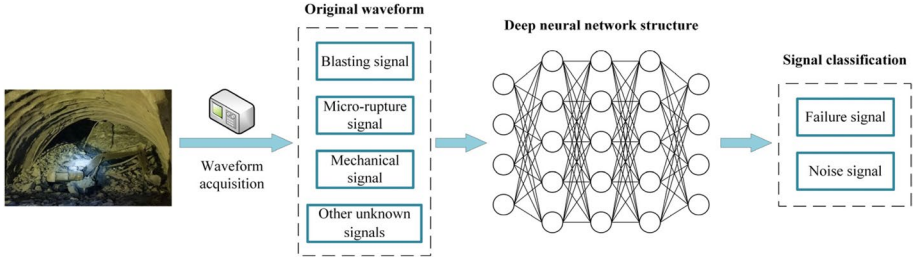


Fig. 6 Schematic diagram of the process of rockburst signal recognition [20]

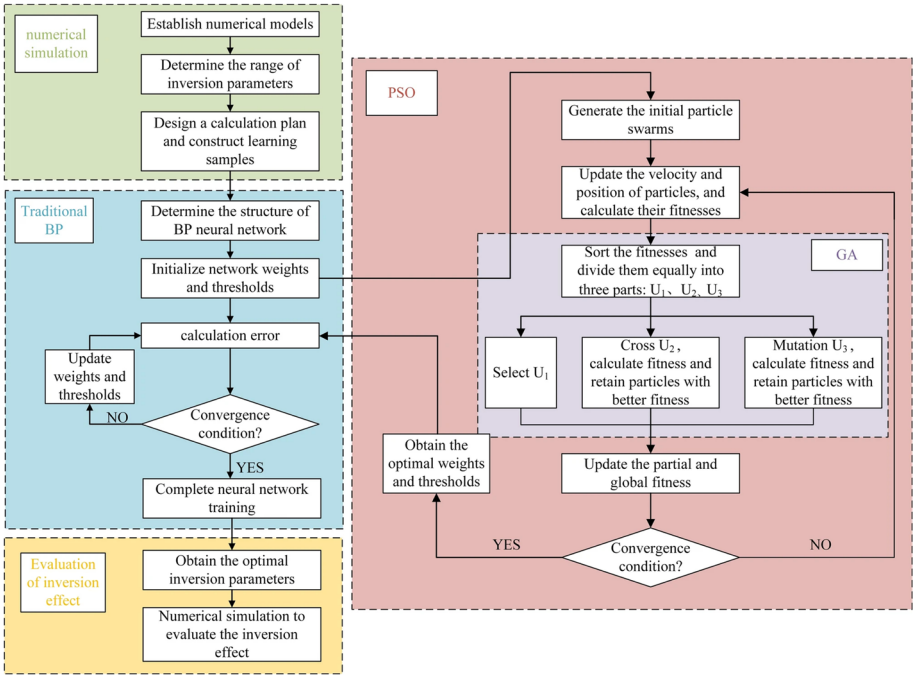


Fig. 7 Intelligent inversion method of rock (body) mechanical parameters [132]

grades [120, 121]. Tunnel face image recognition and classification involves extracting contour edges and joint fracture features from digitally processed images, enabling 3D reconstruction of the tunnel face and automated lithological identification [122–125].

3.1 Parameter inversion analysis

3.1.1 Parameter inversion based on surrounding rock deformation monitoring data

Parameter inversion in geotechnical engineering originated with Kavanagh, who first employed the finite element method to estimate elastic parameters, establishing the foundation for subsequent developments [126]. During the 1990s, the emergence of digital data transmission and resource sharing, as emphasized by Robertson, accelerated the evolution of inversion techniques toward intelligent and automated methodologies [127]. Later, Feng and Hudson expanded this framework by proposing four systematic pathways for intelligent parameter estimation in rock mechanics [128]. As illustrated in Fig. 7, the inversion workflow integrates numerical simulation, hybrid intelligent optimization, and evaluation of inversion accuracy. Initially, numerical models are established to define the inversion parameter range, design computational schemes, and generate

training samples through orthogonal or uniform experimental design using simulation software such as FLAC3D, ABAQUS, or Midas/GTS [129–131]. The generated data train a BP neural network, with its weights and thresholds optimized by a hybrid PSO–GA algorithm. PSO ensures rapid convergence through iterative position and velocity updates, while GA enhances global search ability via selection, crossover, and mutation operations. The optimized parameters are then applied to obtain the optimal inversion results, which are verified through forward numerical simulation to assess predictive accuracy.

In tunnel engineering, inversion analysis primarily targets three parameters: in-situ stress, support structure loads, and geotechnical parameters [133, 134]. Song et al. reviewed progress in these areas, highlighting challenges associated with digitalization and intelligent engineering [135]. While the theoretical framework is mature, current methods suffer from low reliability, lack of objective accuracy metrics, and high computational costs. Research on in-situ stress inversion has advanced with the integration of measurement technologies and DL algorithms, notably those optimized by genetic algorithms for applications in dams [136], mining areas [137], and deep-buried tunnel [138, 139]. Huichao et al. developed a DL algorithm combining continuous physical field features (e.g., temperature, seepage) with discontinuous geological structures (e.g., fault zones) to model in-situ stress fields in complex geological conditions [140].

Similarly, Qian et al. used GAN to optimize lateral stress coefficients and reconstruct the stress field in an underground hydropower station case [141]. In support load inversion, accurate determination of structural loads is essential for safety and stability during construction [135, 142]. Xiangyu et al. combined the PSO algorithm with finite element software for metro tunnel projects [143], though Zou et al. noted challenges in complex rock mass structures or support systems [144]. While DL has seen limited application in support load inversion, the field is less complex than material property inversion due to fewer uncertain variables. Advancements in intelligent monitoring technologies have also enabled more precise measurement of stress relationships between surrounding rock and support structures.

Geotechnical parameters, such as elastic modulus, Poisson's ratio, internal friction angle, and cohesion, are essential for assessing rock mass stability in tunnel engineering. Inversion of these parameters is closely linked to identifying the appropriate constitutive model of the rock mass. DL algorithms have been widely used to train large datasets of these parameters, enabling accurate inversion by capturing nonlinear behaviors of geotechnical media [145–150]. In deep-buried or highly saturated tunnel, complex stress-seepage field interactions due to groundwater activity have led to studies on rheological parameters and creep characteristics under fluid–solid coupling conditions, crucial for design and stability assessment [151–155].

3.1.2 Parameter inversion analysis based on advanced geological prospecting techniques

As tunnel and underground engineering projects extend deeper into subsurface regions, construction faces challenges such as saturated fault zones and karst cavities, which are prone to hazards like collapse, water inrush, and mud inrush [156–158]. These conditions complicate on-site monitoring, but GPR has proven effective for detecting adverse geological bodies ahead of the tunnel face, enabling early warnings of potential disasters and enhancing risk mitigation during excavation [159, 160]. DL has been applied in two

key areas of GPR for tunnel: signal recognition and inversion analysis [161]. GPR data often suffers from irregular defects and steel arch supports that obscure signal reflections, leading to distorted echoes and difficult-to-interpret images [162–165]. To address this, Liu et al. introduced GPR-InvNet, the first DL-based inversion analysis network for GPR data in tunnel [166], and M-YOLACT, a multi-task model for defect segmentation and tunnel lining thickness estimation [167]. Xie et al. developed a U-Net architecture to reconstruct permittivity distributions in real-time using adjacent trace compression and instance normalization [168].

Existing studies often overlook the attenuation of GPR signals with increasing tunnel depth, which limits the performance of models like GPR-InvNet for detecting deep-seated anomalies. To address this, Ji et al. introduced PINet, a DL model that compresses GPR data and extracts depth-dependent features to improve permittivity inversion [169]. Wang et al. developed GPRI-Net, combining CNNs and RNNs for simultaneous permittivity inversion and target signal recognition, enhancing accuracy and speed [170]. Dai et al. proposed DMRF-UNet, a two-stage model that extends 2D GPR data to 3D, using 3DInvNet with multi-scale feature aggregation for better inversion of complex GPR data [171, 172].

The electrical resistivity method is widely used for detecting water-rich structures and identifying water inrush zones due to its cost-effectiveness and sensitivity to groundwater formations [173, 174]. Puzyrev trained FCNs on 1,024,000 synthetic electromagnetic measurements, achieving mean absolute errors below 3% and generalization errors under 5%, demonstrating DL potential for real-time resistivity estimation [175]. Liu et al. developed ERSInvNet, a U-Net-based framework, to address the limitations of conventional resistivity inversion by incorporating deep weighting functions represented in Eq. 1 and smoothness constraints, enhancing sensitivity to deeper structures and overall inversion performance [34, 176, 177]. Subsequent improvements, including refined input–output mappings, weighted constraints, and integration of physics-informed models, further advanced the model's accuracy, efficiency, and applicability to complex geological conditions [178–180].

$$d\left(z\right)=\sqrt{Z+\gamma} \tag{4}$$

In Eq. 4: $d\left(z\right)$ denotes the depth weighting function, which is designed to enhance detection sensitivity in regions near the tunnel face, thereby improving the network's responsiveness to subsurface variations throughout the entire detection process. The depth Z typically ranges from 0 to several hundred meters, depending on the geological conditions, while the weighting coefficient γ is generally between 0.1 and 1.0, allowing for fine-tuning of sensitivity to subsurface changes.

In practical engineering, obtaining measured data using advanced geological prospecting techniques is significantly more challenging than acquiring surrounding rock deformation data [181]. As a result, supervised learning methods like CNNs and RNNs rely heavily on statistical patterns in sample datasets without accounting for the underlying physical laws of geological processes [182, 183]. To address this limitation, Yang et al. proposed an unsupervised DL–based inversion framework that integrates data-driven learning with physical forward modelling [184]. As illustrated in Fig. 8, three inversion workflows are compared: (a) the conventional supervised DL inversion, where a neural network predicts subsurface models by minimizing loss between predicted and true

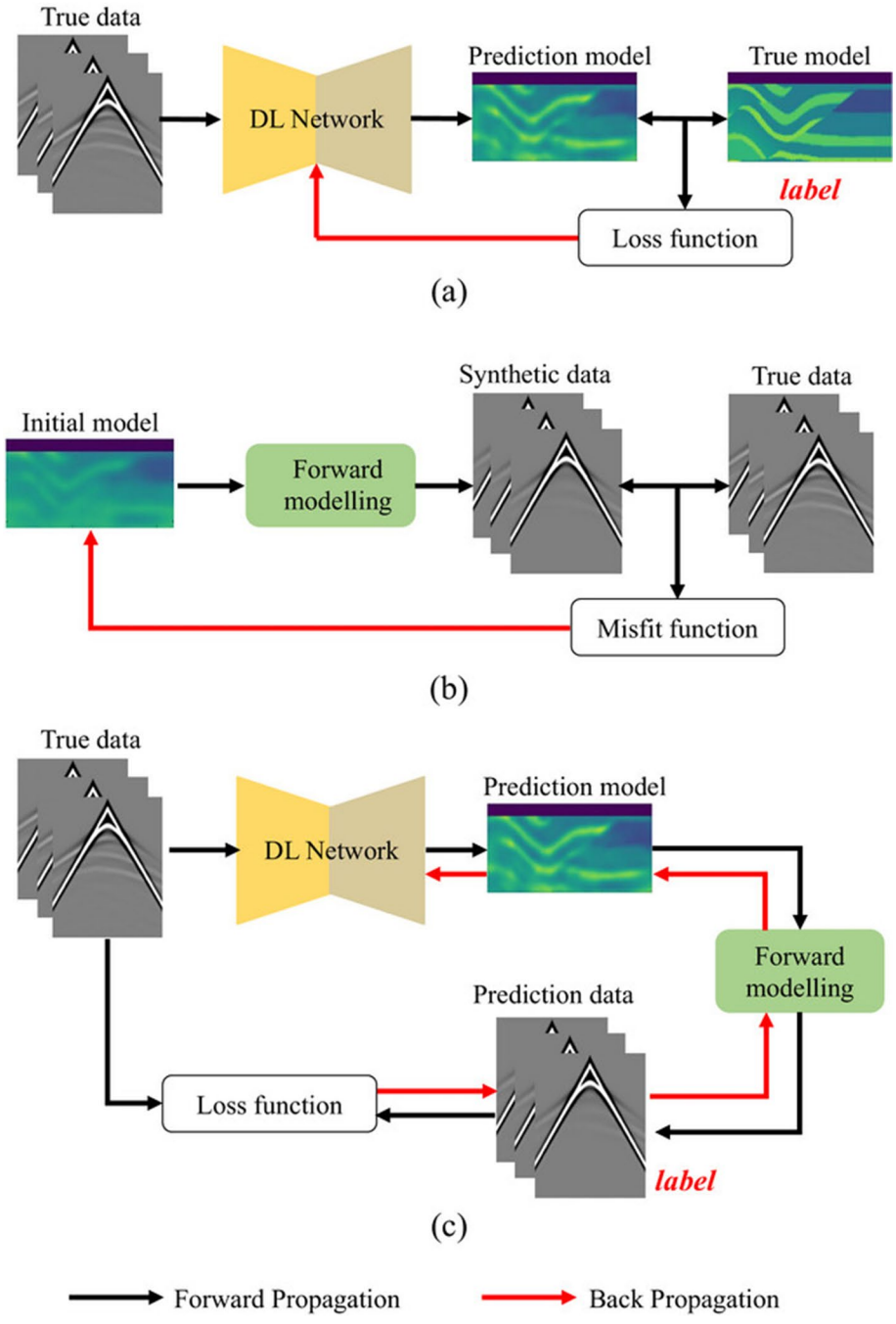


Fig. 8 Workflows of **a** supervised DL inversion, **b** physics-based forward modelling, and **c** the proposed unsupervised hybrid inversion integrating DL prediction and forward modelling through a joint loss–misfit function [184]

models; (b) the physics-based forward modelling approach, which iteratively updates an initial model by minimizing the misfit between synthetic and observed data; and (c) the proposed hybrid unsupervised method, which couples data-driven prediction with forward modelling through a joint loss–misfit function, enabling self-optimization without labeled ground truth. This unified framework enhances inversion accuracy, robustness, and physical interpretability, particularly when measured field data are limited.

A large synthetic dataset was generated for training, including 12,000 two-dimensional overthrust models and 10,000 random velocity models with velocities ranging from 1500

to 5500 m/s. Seismic forward modelling using 20 sources and 100 receivers at a 15 Hz dominant frequency produced the training data. A multi-scale strategy, incorporating low-, mid-, and high-frequency components, further improved inversion precision and mitigated cycle-skipping, outperforming conventional Full Waveform Inversion (FWI) techniques [184].

Based on the above analysis, it can be concluded that DL has significantly enhanced parameter inversion analysis by improving accuracy, efficiency, and the automation of data processing and interpretation. It effectively integrates multisource datasets and allows for dynamic learning and updates of training data. Future research should incorporate uncertainty-aware inversion frameworks, such as probabilistic neural networks and Bayesian DL, to quantify confidence intervals for predicted geotechnical parameters under variable stress–strain conditions. Furthermore, applying these methods to diverse tunnel projects is crucial for validating their feasibility and effectiveness in real-world engineering scenarios.

3.2 Prediction and optimization of construction machinery parameters

3.2.1 Prediction of excavation parameters considering spatiotemporal characteristics

A summary of studies on excavation parameter prediction is provided in Table 2. As indicated, the sample size required for predicting excavation parameters is significantly larger than that needed for parameter inversion- typically exceeding 1,000 samples, and in some cases, reaching into the tens of thousands. Therefore, it is necessary to perform data preprocessing to identify and retain features with high-dimensional information content [185]. Commonly used methods include Principal Component Analysis (PCA) and Pearson correlation analysis [186]. The selection of input and output parameters is not fixed; input parameters generally fall into two categories: geological parameters and machinery parameters, while the most commonly used output parameters are the cutterhead rotational speed and torque [187–189]. It is particularly important to note that when predicting the movement trajectory or operational status of excavation machinery, the model's output parameters represent key control indicators for machine actuation [190, 191]. These primarily include horizontal and vertical deviations at the front and rear ends of the machine, as well as its attitude information, such as roll and pitch angle [192, 193].

Further analysis of Table 2 reveals that the LSTM network is one of the most widely used DL algorithms for tunnel parameter prediction tasks. While LSTM is widely used for sequential data due to its ability to capture long-term dependencies, simpler models like 1D-CNN and traditional ML approaches such as SVM or Random Forests can offer comparable performance when the data complexity is lower or less sequential in nature. These simpler models typically require less computational power, offer faster training times, and are more interpretable. The choice between LSTM and these alternatives depends on the task complexity and the nature of the data; for tasks without significant temporal dependencies, traditional ML models or CNNs might be preferred due to their efficiency and ease of implementation [212–214].

This is mostly because, during the excavation process, construction machinery is equipped with numerous sensors that monitor and collect extensive data related to equipment parameters, geological conditions, and fault indicators, which typically exhibit strong temporal dependencies [215, 216]. LSTM is an improved RNN designed

to overcome the vanishing and exploding gradient problems in traditional RNNs. Unlike RNNs, which use a single hidden state, LSTM incorporates memory cells (Ct) and three gates: forget gate (ft), input gate (it), and output gate (ot). These gates control the flow of information, allowing LSTM to selectively forget or update past states based on new inputs, making it effective for learning long-term dependencies in sequences [217–221]. A schematic of the LSTM structure is shown in Fig. 9.

Studies have found that in the prediction of tunnel equipment excavation parameters, it is essential to consider historical data and also to utilize future data from subsequent excavation processes. To address this challenge, studies have employed the Bi-LSTM neural network for real-time prediction of tunnel parameters. Additionally, attention enhancement mechanisms have been introduced to improve the interpretability and robustness of the model, thereby providing more reliable decision support for intelligent tunnel operations. Although DL algorithms can achieve high prediction accuracy on certain datasets, their performance may degrade significantly in complex geological conditions or when the dataset contains a large amount of noisy data. Moreover, as the number of neural network layers increases, issues such as vanishing gradients and model degradation may arise during training. To address these challenges, Hu and Cheng proposed a hybrid neural network model that combines an Attention mechanism, a CNN-based feature extractor, and an LSTM-based predictor for the purpose of predicting the excavation rate of shield tunnel machines.

Studies have emphasized the need to consider both historical and future data for predicting tunnel equipment excavation parameters [223–225]. To address this, Bi-LSTM network has been employed for real-time prediction [226–229]. Attention mechanisms have also been integrated to enhance model interpretability and robustness, aiding intelligent tunnel operations [230, 231]. However, DL performance can degrade in complex geological conditions or with noisy data [206, 232]. Additionally, increasing neural network layers can lead to vanishing gradients and model degradation [233, 234]. To mitigate this limitation, UQ and cross-validation techniques-such as Monte Carlo dropout, ensemble calibration, and k-fold cross-validationshould be systematically integrated to evaluate model robustness and mitigate overfitting risks in real-world tunnel scenarios. Hu and Cheng proposed a hybrid model combining attention mechanisms, CNN-based feature extraction, and LSTM-based prediction for shield tunnel excavation rate prediction [235].

3.2.2 Dynamic performance optimization and real-time control of tunnel boring machines

China has become a global leader in highway and railway tunnel construction, while the rapid urbanization has greatly accelerated the development of urban rail transit systems [236, 237]. However, research on the relationship between various types of tunnels and the performance of their corresponding tunnel machinery remains relatively lagging behind. Shield tunnel machines and road headers are among the most frequently used equipment in tunnel and underground construction [238]. Yet, their performance regulation and adjustment still predominantly rely on expert judgment, rather than data-driven or intelligent control systems. Jia et al. illustrated the Optimal Attitude Control Policy framework as shown in Fig. 10, combining real-time Geological Environment Classification data, TBM, Tunnel Boring Machine Attitude and Position (TBMAP)

Table 2 Statistics of literature on the prediction of tunnel parameters

Deep learning algorithm	Input parameters	Output parameters	Sample size	Independent validation	Computational requirements	Geographic location/tunnel type	Refs.
Temporal Convolutional Network (TCN)	Cutterhead rotation speed, penetration rate, cutterhead torque	Cutterhead torque, total thrust, TBM advance rate	4,540 Sects. (7:2:1 split)	Yes	GPU-based (RTX 3080Ti, PyTorch, Ubuntu 20.04)	Yinchao project, inner mongolia, China — Open-tsype water diversion tunnel	[194]
Artificial Neural Networks (ANN)	Uniaxial Compressive Strength (UCS), Point Load Test (PLT) (I50)), Rock Mass Rating (RMR), Rock Quality Designation (RQD), Porosity (n), Brazilian Tensile Strength (BTS)	Penetration Rate	235	No (tenfold cross-validation only)	Low (MATLAB-based; no GPU used)	China — Water conveyance tunnel, Xianglushan Project (granitic rock)	[195]
Multi-Channel Decoupled Deep Neural Network Model	Advance rate, cutterhead speed, geological type	Thrust and torque	1,292	Yes — validated on an independent lower-tunnel dataset (Test Set 2)	Implemented in Python (Keras + Sklearn); moderate computing — model training with grid search; no GPU details specified	Singapore — Stacked twin-bored tunnel (North–South Corridor, Shenton Way–Maxwell Road) using Earth Pressure Balance TBM	[196]
Long Short-Term Memory (LSTM) Network	Thrust, cutterhead torque, cutterhead rotation speed, earth pressure	Advance rate	14,000	Yes — Testing on 20% unseen data; cross-validation with multiple time steps	Moderate (TensorFlow/Keras, CPU-based; no GPU specified; 30 epochs, batch size 10)	Singapore — Thomson–East Coast Line (T308), EPB TBM in marine clay	[197]
	UCS of rock mass, tensile strength, slope coefficient, etc	Advance rate	1,125	Yes — 80% training, 20% independent test	Moderate — implemented in Python (TensorFlow); Intel i7 CPU @2.60 GHz, 32 GB RAM; no GPU used	Iran — Alborz Service Tunnel, Tehran–Shomal Motorway (6.4 km, open-type gripper TBM in hard rock)	[53]
Deep Q-network (DQN)	Normalized soil properties (unit weight, cohesion, friction angle, Young's modulus), normalized overburden depth	Optimal tunnel face support pressure (50, 100, 150, 200, 250 kPa)	400 training episodes	Yes — transfer learning from analytical to numerical environment; validated on unseen random geologies	Moderate–High: Implemented in Python (PyTorch); trained for 400 episodes; retraining required ~ 30 min/episode; numerical simulations via FLAC3D	Simulated TBM operation (no field dataset) — 10 m diameter tunnel in soft soil; modeled for EPB/SPB shield types	[198]
GRU Neural Network	Construction parameters such as force, penetration rate, grouting pressure, and grouting volume	Cutterhead torque	1,000	Yes — 80% training, 20% testing; ten-fold cross-validation used	Moderate — Python-based (Adam optimizer); no GPU specified; 150 iterations	Shenzhen International Airport Tunnel, China — EPB shield tunnel (soft–hard mixed strata, 3.3 km length)	[199]

Table 2 (continued)

Deep learning algorithm	Input parameters	Output parameters	Sample size	Independent validation	Computational requirements	Geographic location/tunnel type	Refs.
Improved PSO-BP Neural Network	Lithology, surrounding rock classification, and groundwater activity	Advance rate in stable excavation sections, total thrust, and cutterhead torque	4,463 TBM working cycles (7:2:1 split)	Yes — independent validation and test set	Moderate (Python, Adam optimizer, 100 epochs, batch size 64; no GPU mentioned)	Jilin Province, China — Silver Pine Project (mixed strata, hard rock TBM)	[200]
Novel DL Approach (Causality-Guided Explainable Model)	Parameters determined through causal-based feature selection	Ground settlement risk	177 monitoring datasets (train/test split 7:3)	Yes — 70% training, 30% testing; multiple sample-size robustness validation	High: Intel i9-13900KF CPU, NVIDIA RTX 4070Ti GPU, 64-bit Windows 11; 1,125 s for hyperparameter grid search	Wuhan, China — San-yang Road Tunnel (Rail-cum-Road Yangtze River Cross Tunnel, 15.2 m diameter, 2.59 km length)	[201]
PSO-based Bidirectional Long Short-Term Memory (Bi-LSTM)	Geological (MSPT, Et, Eb, Kt, Kb), Geometric (Cover depth, MTC, HD, VD), Operational (Thrust, Cutterhead rotational torque, Cutterhead rotational speed, Excavation rate, Penetration rate, Grouting pressure, Chamber earth pressure)	Maximum ground settlement (Sp)	126 datasets (100 training, 26 test)	Yes — independent test set used	Moderate — implemented in Python (TensorFlow 2.8); trained on NVIDIA RTX 3060 GPU; 200 epochs	Nanjing, China — Metro Line 7 EPB Shield Tunnel (soft clay and sand strata)	[202]
Random Forest (RF) — compared with Logistic Regression, SVM, BP Neural Network	Shield tunnelling chamber temperature, average excavation speed, liquidity index, plasticity index, cutterhead torque, total thrust force	Mud cake formation	4,525 data samples (80% training, 20% testing)	Yes — independent test set (905 samples); evaluated via precision, recall, F1-score	Moderate — Python 3.10, Sklearn; 10 input features; 1,000 training iterations; CPU-based	Fuzhou, China — Binhai Express Metro, EPB shield tunnel in clayey strata (soft–hard mixed ground)	[203]
Extreme Gradient Boosting (XGBoost)	Tunnel burial depth (H), tunnel diameter (D), strength-to-stress ratio (SSR), support stiffness (K)	Tunnel squeezing	7,538 data samples	Yes — 70% training, 30% independent testing	Moderate — Python (Sklearn, LightGBM, XGBoost); trained on CPU (i7-11700H, 32 GB RAM); 500 iterations	Chongqing, China — Metro Line 9, double-track EPB shield tunnel through mixed gravelly strata	[204]

Table 2 (continued)

Deep learning algorithm	Input parameters	Output parameters	Sample size	Independent validation	Computational requirements	Geographic location/tunnel type	Refs.
CNNs and U-Net	Random field images (reflecting soil parameter variabilities)	Critical Support Pressure, Collapse Modes (Failure Modes)	450 training samples	Yes — 70/30 train-validation split; independent Monte Carlo dataset for testing	Moderate — MATLAB DL Toolbox; single GPU (NVIDIA RTX 3060); 35 min training, 30 s prediction (vs. 18 h for FELA)	Numerical case study — Layered soil tunnel model (two-layer random field system; synthetic tunnel, D = 10 m, C = 10 m); not based on a specific geographic tunnel	[205]
Hybrid Neural Network	Advance rate, cutterhead torque, cutterhead rotation angle	Horizontal and vertical deviations of the shield head and shield tail	5,005	Yes — 80% training, 20% testing split; independent test dataset used for validation	Moderate — Implemented in Keras + TensorFlow, Intel i5–4590 CPU, NVIDIA GTX950 GPU (8 GB RAM), 7 s per epoch × 200 epochs	Wuhan, China — Yangtze Sanyang Road Tunnel, Mixshield TBM, 15.76 m diameter, 2.59 km river-crossing tunnel (soft–hard mixed strata, high-pressure conditions)	[206]
Hybrid Deep Neural Network	Earth pressure, jack speed, cutterhead torque at the current time step	Cutterhead torque at the next time step	60,000	Yes — multiple independent test splits (7:3, 8:2, 9:1)	Moderate — Implemented in Keras/TensorFlow, trained on Intel i5-10210U CPU, 2 GB GPU, 100 epochs, learning rate = 0.002	Singapore — Metro T225 Project, EPB shield tunnel, 6.68 m diameter, 750 m length (501 rings)	[207]
Hybrid DL Approach (GCN and LSTM)	Spatial and temporal features from TBM parameters and geological information	Advanced geological conditions (rock types, water intrusion, karst caverns, surface subsidence)	28,000	Yes — progressive validation (9 segments with rolling training/testing sets mimicking real-time tunnelling)	Trained on workstation (CPU i5-12600KF, GPU RTX-3070, 32 GB RAM); 30 epochs, batch size = 100; implemented in Python (TensorFlow/Keras)	Singapore — Thomson–East Coast Line (Marine Terrace ↔ Siglap Station), EPB-TBM, 6.67 m diameter, 1.33 km tunnel, 954 rings	[23]
BPNN/Permutation importance-based BPNN (PI-BPNN)	Tunnel depth, distance between tunnel face and surface monitoring points (DTM), weighted average soil compressibility modulus (ACM), grouting pressure, penetrating rate, thrust force	Tunnel-induced ground settlements	187 samples; 14 input variables	Yes — 70% training, 30% testing split; repeated 1000 × for variance/stability analysis	Implemented in MATLAB + Python (Scikit-learn); standard CPU computation; moderate complexity	Yuji Tunnel, China — Slurry Shield TBM, 12.4 m diameter, 31–36 m depth, mixed silty clay & sand ground	[208]
Two-dimensional CNN (2D-CNN)	Soil Young's modulus random field	Horizontal and vertical convergences of high-speed railway tunnel	15,000 training samples	Yes — 12,000 for training, 3,000 for independent testing	Moderate–High: Implemented in Python (TensorFlow); trained on GPU (RTX 3090, 24 GB); 500 epochs, batch size = 64	Simulated case — High-speed railway tunnel in spatially variable soils (no specific geographic tunnel)	[209]

Table 2 (continued)

Deep learning algorithm	Input parameters	Output parameters	Sample size	Independent validation	Computational requirements	Geographic location/tunnel type	Refs.
Graph Attention Networks (GAT) with GRU, GAT-GRU	Stress, Strain, Temperature, Deformation (Displacement) at multiple nodes, from past time steps	Future deformation data (stress, strain, temperature, and displacement)	1,440 hourly data points (from 14 monitoring nodes, Jan–Mar 2024)	Yes — 70% training, 30% testing; grid search for optimal hyperparameters	Moderate — Python (PyTorch), RTX GPU, 150 epochs, batch size = 64, learning rate = 0.01	Shandong Province, China — operational highway tunnel with dense multi-sensor layout (14 nodes, 2 m spacing)	[210]
One-dimensional CNN (1D-CNN)	Spatial distribution of Young's modulus across tunnel segments	Transverse deformation and vertical deformation of the segments	2000 deformation samples	Yes — 80% training, 20% test	Moderate — Python (PyTorch 2.5.1), Intel i7-12700U CPU, 32 GB RAM; 1000 epochs, batch size 64; RMSE & R^2 monitored for convergence	Simulated shield tunnel segments (outer Ø 6.2 m, inner Ø 5.5 m) — no specific geographic tunnel; based on Chinese metro-type section	[211]

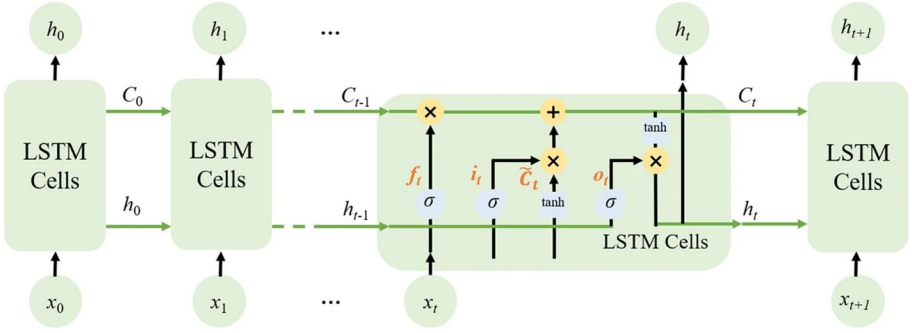


Fig. 9 Schematic diagram of LSTM structure [222]

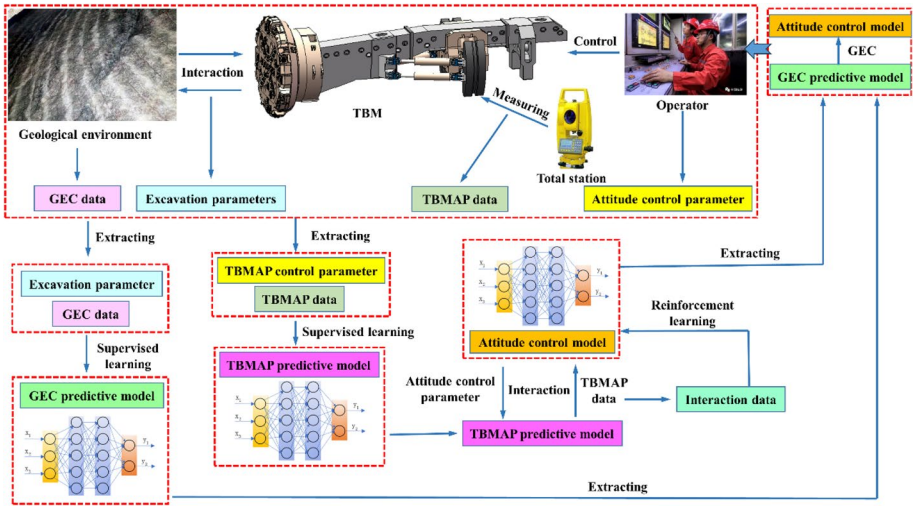


Fig. 10 Framework for real-time optimal attitude control of TBM [239]

parameters, and RL to optimize TBM attitude control, minimizing deviation from the designed tunnel axis [239].

Predicting failures in tunnel equipment is critically important to ensure effective performance control during construction [240]. To this end, researchers have employed various DL algorithms to explore performance optimization and control strategies for excavation machinery. Fu et al. combined LSTM networks with GNNs to dynamically predict the operating posture and position of tunnel machines during excavation [241]. The study further enhanced the model's interpretability using Shapley Additive Explanations (SHAP). In contrast, Pan et al. improved a DL model combining Graph Convolutional Networks GCNs and a scaled dot-product attention mechanism to predict TBM performance. The *att-GCN* model achieved an MAPE of 15.475% for the penetration rate and 15.173% for the energy consumption, improving accuracy and interpretability compared to traditional algorithms [242].

Gao et al. proposed a real-time earth pressure control model for shield tunnel based on DL by integrating the GRU network with a genetic optimization algorithm [243]. The model employs a Relative Error-Based Cost Function (REMSE) as shown in Eq. 5 as its loss function, which helps average the gradients across different samples, thereby accelerating the training process [244]. Elbaz et al. identified geological conditions, tunnel machine operating parameters, and geometric configurations as the three primary

factors for designing excavation machinery. By combining a CNN with an LSTM network enhanced by clustering fusion algorithms, they developed the KCNN-LSTM DL model, which is used to predict and control the energy consumption of the cutterhead drive system in tunnel machines [245].

$$J_{REMSE}^{0} = \frac{1}{M}\sum_{i=1}^{M}\left(\frac{y_{oi}-y_i}{y_i}\right)^2 \tag{5}$$

In the Eq. 5: J_{REMSE}^{0} represents the RE loss function; M denotes the number of training samples; y_{oi} is the predicted value of the model for the i-th sample; y_i is the target value for the i-th sample.

Studies have explored integrating Deep Reinforcement Learning (DRL) with physical information fusion to improve autonomous excavation in TBM [246, 247]. This combination leverages prior physical model knowledge with adaptive learning, addressing traditional DRL limitations like low learning efficiency and poor predictive accuracy. For improving automation in coal mine roadways, an intelligent cutting strategy combining LSTM networks with fuzzy inference control has been proposed to optimize cutting head speed and arm swing for better excavation efficiency and precision [248]. Additionally, seismic data in tunnel construction, often affected by noise, has been denoised using a U-Net architecture with pre-processing techniques, enhancing model generalization through transfer learning [249]. Liu et al. [250] and Fu et al. [251] also applied DL with multi-objective optimization to predict and control TBM posture and position, leading to improved machine performance in complex conditions.

DL has advanced in optimizing tunnel machine performance, particularly by using CNNs and RNNs to process excavation data [252]. This allows for the prediction of cutterhead wear and real-time adjustments to construction strategies, reducing downtime and improving excavation efficiency. Kilic et al. developed a 1D CNN for predicting cutter wear in Pipe Jacking EPB TBMs, achieving an R^2 of 89.6% and a computation time of just 3 min 22 s. Their model efficiently predicts wear in real-time, offering a fast, reliable solution for optimizing TBM performance and improving maintenance scheduling, particularly in soft ground tunnel. This highlights the potential of DL models as effective alternatives to traditional methods in mechanized tunnel [54]. DL also helps adjust excavation force and direction, ensuring accurate tunnel alignment and enabling real-time performance monitoring. Future advancements, especially with 5G and remote-control technologies, will integrate geological data, sensor inputs, and DL for autonomous decision-making, intelligent monitoring, and energy-efficient tunnel operations.

3.3 Construction and operation process control with risk assessment

3.3.1 Tunnel construction decision support based on multisource data fusion

The NATM and the Drill and Blast Method have become the primary construction techniques for highway and railway tunnel projects due to decades of experience and refinement [253]. NATM relies on surrounding rock classification and maximizes the self-supporting capacity of the surrounding rock and the restraint effects of the tunnel face [254]. Incorporating on-site monitoring data enables timely adjustments to surrounding rock and support structure parameters during excavation, as well as optimal timing for lining installation [255]. In contrast, the Drill and Blast Method requires

precise selection and prediction of blasting parameters, as they directly impact the stability of the support structure, excavation speed, and construction safety [256]. A major challenge in today's engineering work is combining data from many sources-site monitoring, lab tests, theory, and computer simulationsto support smart, data-based decisions on construction sites. [257–259]. Wu et al. proposed a method to improve tunnel collapse risk assessment by fusing multi-source data using the Improved Dempster–Shafer Theory and AI models [46]. Figure 11 outlines the process where data from geological forecasts, site inspections, and monitoring are analysed using Cloud Models (CM), Gradient Boosting Decision Tree (GBDT), and SVM. A weight coefficient is adjusted based on the conflict coefficient (K) to fuse evidence, providing a more accurate and reliable collapse risk prediction.

Traditional neural network design in construction decision-making can be challenging due to constraints in determining hidden layers, nodes, and activation functions, while

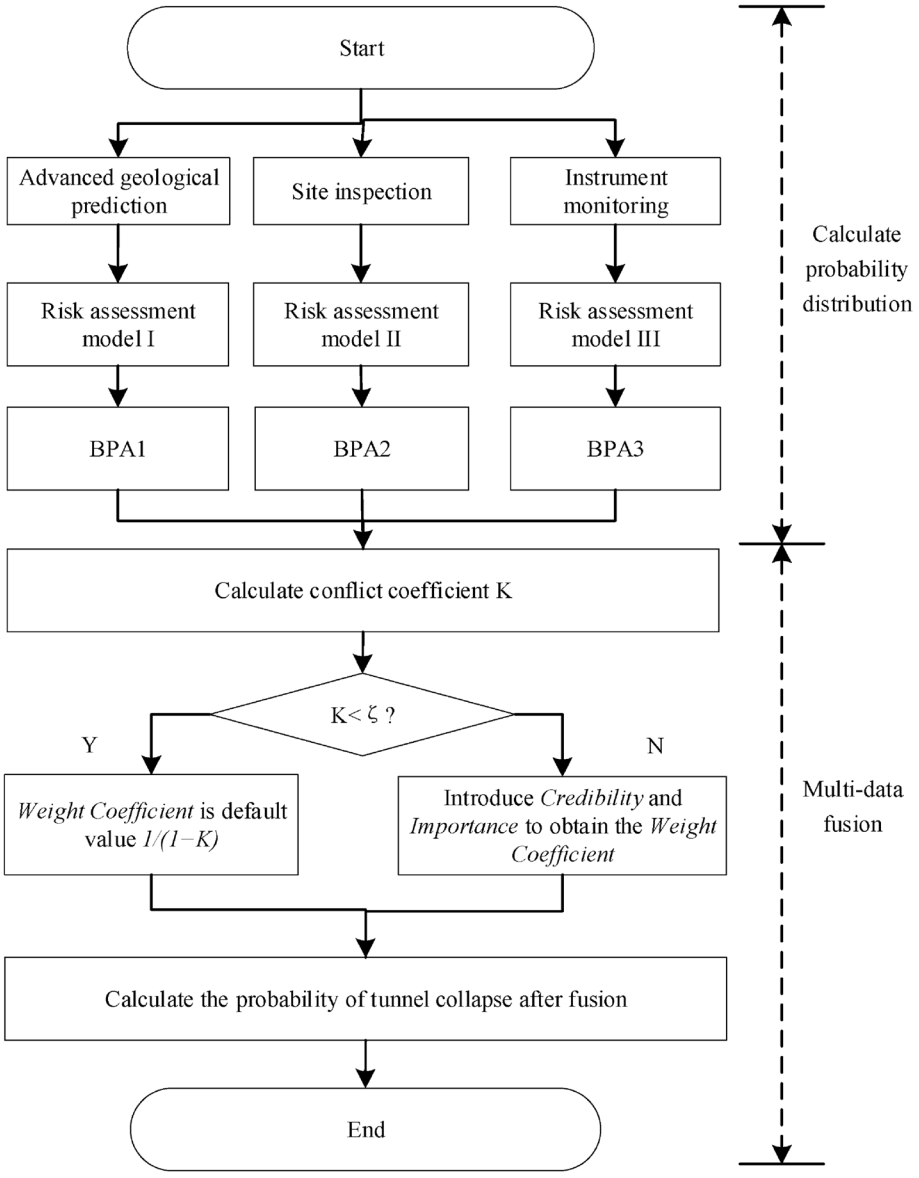


Fig. 11 Flowchart of the proposed hybrid method for multi-source data fusion decision [46]

conventional data processing methods impact decision timeliness and model generalizability [260]. Advancements in AI, particularly DL, enable more objective and intelligent decision support [261]. Wu et al. used a CNN to extract tunnel monitoring video data, integrating domain knowledge from NATM with a Bayesian model to automate construction progress identification, addressing DL "black-box" challenge [55].

Erharter et al. proposed that tunnel excavation involves multiple sub-processes, which require simplification for effective simulation. The study developed a RL-based tunnel strategy to optimize the construction process, with the goal of tunnel breakthrough [262]. Different excavation sequences are evaluated using cumulative reward scoring to guide strategic planning. A schematic of this RL-based strategy is introduced by Elbaz et al. as shown in Fig. 12. Another study by Zhang et al. (2023) introduced a cross-platform deployable tunnel robotics system featuring a closed-loop intelligent control architecture. The system, using a PSO algorithm, minimizes energy consumption per excavation unit, supporting energy conservation and emissions reduction in tunnel construction [263].

With the advancement of mechanical equipment technologies, high-precision drilling systems and blasting techniques have significantly improved the accuracy and efficiency of tunnel and underground construction [265]. These innovations have not only optimized construction workflows but also reduced the negative impact on surrounding environments. At the same time, the widespread deployment of CCTV monitoring and acquisition systems has greatly enriched the variety of on-site monitoring data [266, 267]. The video and image data captured by these systems play a crucial role in tracking construction progress, implementing safety management measures, and conducting post-construction quality assessments [268, 269]. Furthermore, the application of DL enables integrated analysis of surrounding rock parameters, construction parameters, and monitoring information, making data processing more efficient and accurate, and significantly enhancing decision-making capabilities in tunnel construction management [82, 230, 270, 271].

Based on the above analysis, many researchers have developed visualized intelligent construction decision support systems to acquire real-time data for dynamically adjusting design plans and optimizing construction processes. This real-time, data-driven dynamic adjustment mechanism enables construction teams to effectively respond to unforeseen conditions in underground engineering projects, thereby ensuring the safety, reliability, and cost-effectiveness of the overall project. Therefore, the integration of such

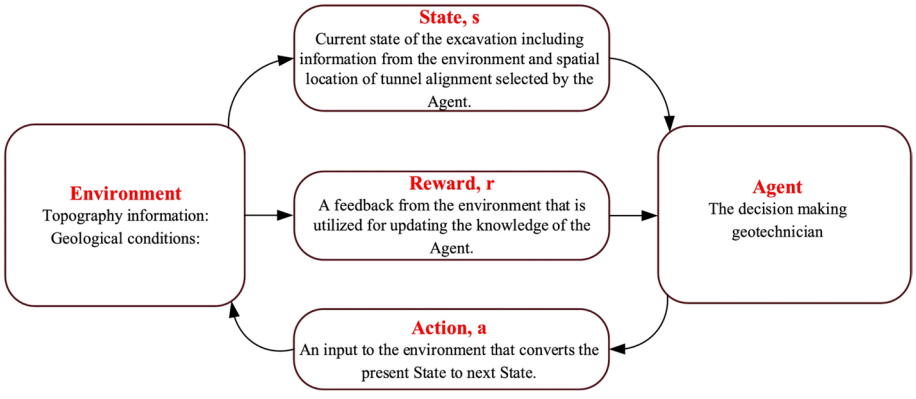


Fig. 12 Schematic of tunnel construction strategy based on reinforcement learning [264]

technologies plays a critical role in supporting the dynamic design and digital construction of tunnel and underground infrastructure, marking a significant step toward the intelligent and refined transformation of construction management practices.

3.3.2 Dynamic risk assessment during tunnel construction and operation phases

The Fourth Industrial Revolution has significantly advanced AI technologies and ushered in the era of big data, prompting the emergence of data-driven modelling for risk control and assessment in tunnel and underground structures during the operational phase [272, 273]. Traditional tunnel risk evaluation methods rely heavily on expert interpretation of model outputs, which limits their applicability across varying geological and environmental conditions, resulting in low reliability [274–276]. Additionally, the nonlinear nature of risk control and evaluation, combined with multiple-source variables, makes it challenging to obtain accurate prior knowledge in real-world scenarios, undermining the effectiveness of risk assessment models [277–279]. A schematic diagram of risk occurrence mechanisms during construction and operation phases is shown in Fig. 13.

DL algorithms are increasingly recognized for their ability to analyse large, high-dimensional data and reveal hidden patterns, making them effective for prediction and decision-making tasks [283–285]. They excel in processing complex, multisource, and time-sequenced monitoring data. By extracting deep features and providing real-time dynamic data, DL enhances accuracy and deepens understanding of internal structural behavior, supporting risk assessment in tunnel and underground operations. Research primarily focuses on two areas: (1) Structural safety models during operations [286–288], such as optimization for tunnel safety, subway evacuation, and lighting energy monitoring in tunnel; and (2) Risk assessment of construction impacts on adjacent tunnel [277, 289, 290] including stability and sensitivity analysis.

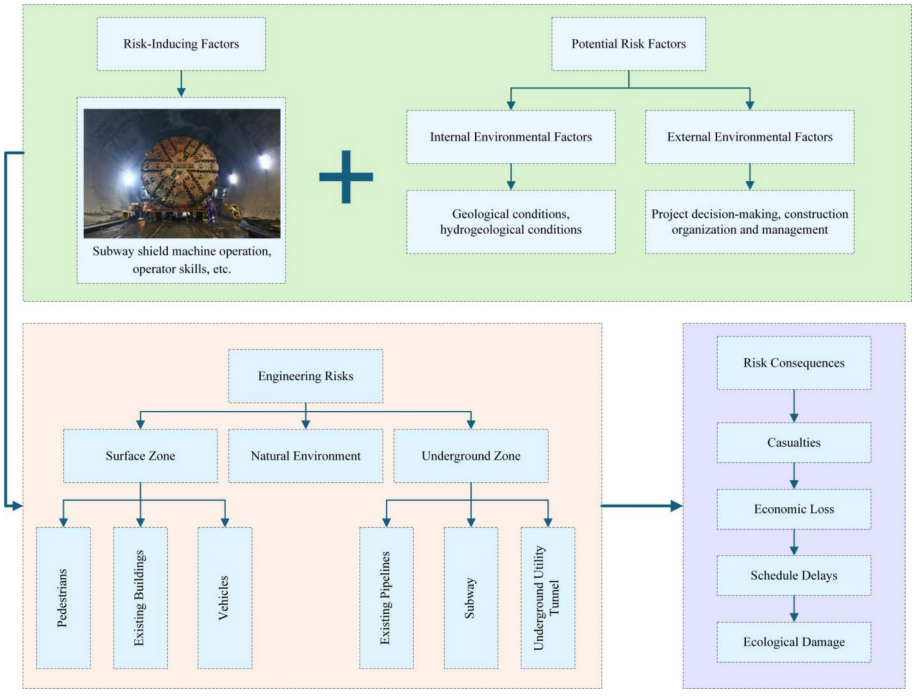


Fig. 13 Schematic of risk occurrence mechanism during construction and operation [280–282].

DL can analyse historical monitoring data (e.g., displacement, stress–strain) during tunnel construction and operation to assess structural health and track personnel and vehicles [291–293]. It can predict structural damage evolution, providing early warnings for hazards like collapses [113], deformations [294], and coal mine accidents [295]. Additionally, DL can: detect vehicles and traffic anomalies in surveillance video [296]; assess tunnel environmental conditions from sensor data (air quality, temperature, humidity); and integrate with Non-Destructive Testing (NDT) for automated pipeline detection and infrastructure inspection [297, 298].

The data-driven technology framework, shown in Fig. 14, highlights DT as a key component. Using point cloud data and 3D laser scanning, DT creates high-precision digital models that replicate the internal 3D spatial reality of structures [299, 300]. In 2013, Lee et al. developed a cloud-based DT platform for real-time prediction and analysis of industrial systems [301]. Latif et al. created a tunnel excavation rock-machine DT framework with five components: physical entity, virtual model, data repository, system services, and connectivity layer [56]. They integrated DL to map rock and machine parameters, developing a virtual model for predicting surrounding rock classifications.

In recent years, the integration of DL and DT technologies has gained traction in tunnel engineering, covering areas like tunnel lining defect detection, full life-cycle platform development, and structural maintenance [138, 303]. Embedding DL within DT frameworks has led to the development of intelligent systems for efficient decision-making, autonomous sensing, and real-time interpretation, significantly improving planning, design, monitoring, and management in tunnel projects [230, 304]. While traditional methods like the NATM and Drill and Blast remain common, increasing project complexity necessitates advanced approaches. DL, through multisource data fusion and intelligent decision support, enables real-time monitoring data processing, optimizing

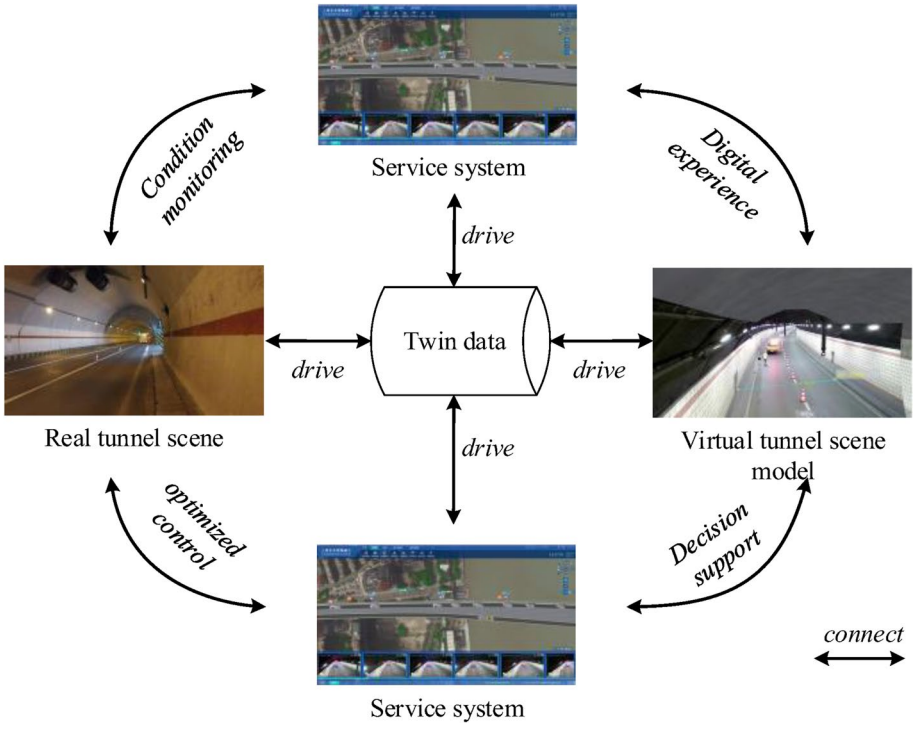


Fig. 14 Data-driven technology [302]

tunnel operations and structural adjustments [305, 306]. This integration of DTs and DL is advancing tunnel construction toward greater automation, intelligence, reliability, and efficiency [307, 308].

Another critical challenge during tunnel construction is concept drift, where geological conditions can unexpectedly change, reducing model accuracy. DL models often assume stable conditions, but when faced with dynamic changes, their performance can degrade. To manage this, techniques like online learning and transfer learning can be applied to adapt models in real-time, allowing for continuous updates as new geological data becomes available. Future research should focus on adaptive frameworks that can accommodate such fluctuations, ensuring model robustness in unpredictable environments.

3.4 Tunnel safety monitoring and defect detection

3.4.1 Identification and detection of surface defects in tunnel lining structures

Tunnel lining structures face various external stresses, such as surrounding rock weight, cyclic loads, groundwater erosion, and fatigue, which can lead to cracks, spalling, and water seepage [309, 310]. These defects threaten stability, reduce lifespan, and endanger train operations and passenger safety [20]. Regular defect detection is necessary, but manual inspections are hindered by factors like poor lighting, dust, gases, and inspector subjectivity, leading to inefficiencies and inaccuracies [311–313].

Early detection methods used edge detection, mathematical morphology, and traditional ML [314, 315], but these were prone to misclassification due to reliance on manually annotated features. As tunnel construction increases, so does the need for efficient defect detection [316]. DL algorithms, which process large datasets, provide greater accuracy in challenging conditions and outperform traditional methods [27, 138]. For example, a dual GAN framework enhances GPR image quality through bidirectional feature extraction and adversarial learning as shown in Fig. 15. Zhu et al. proposed an intelligent system using Mask R-CNN for more efficient tunnel inspections [57], and Xiong et al. integrated AI and computer vision for multisource data fusion in high-speed railway tunnel, improving automation and reliability [317].

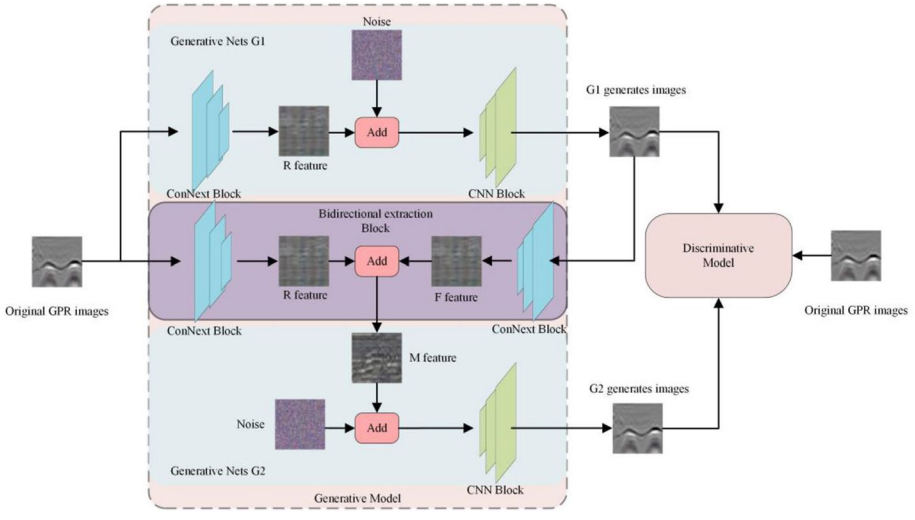


Fig. 15 Deep learning-based dual-GAN model for enhancing GPR images in tunnel defect detection [318]

The integration of DL, NDT, and sensor data acquisition has become the dominant approach for tunnel defect monitoring, enhancing accuracy, automation, and efficiency [319]. The focus of this study is on detecting cracks and water seepage, key indicators of structural deterioration, which are prioritized in the proposed intelligent detection framework [320].

3.4.1.1 Intelligent detection of crack damage Cracks in tunnel structures reduce stiffness and load-bearing capacity, potentially evolving into through-cracks that compromise safety and stability [321, 322]. DL algorithms, utilizing GPU-accelerated image processing and CUDA parallel computing, are highly effective in detecting cracks and water seepage [323]. Zhang et al. (2016) used CNNs for precise crack classification and localization through feature extraction and pixel-wise probabilities [324].

Yang et al. developed a hybrid DL framework [325], combining Faster R-CNN for region-level localization and U-Net for pixel-wise segmentation [326–328]. The Faster R-CNN's Region Proposal Network (RPN) detects cracks, while U-Net ensures accurate pixel-level segmentation by integrating shallow and deep features. DL excels in crack detection due to noise suppression, hierarchical feature extraction, and fine-grained classification. A schematic of the U-Net architecture is shown in Fig. 16.

Crack detection research has focused on enhancing DL algorithms for greater accuracy and efficiency, with many studies exploring ways to adapt these models to different construction conditions. Xu et al. (2024) proposed an automatic tunnel crack detection method using a mobile image acquisition system combined with an enhanced DL ensemble model like YOLO-LD. Their approach improved detection accuracy, speed, and adaptability, with field tests demonstrating increased precision and fast inference times, confirming the robustness of their solution [330].

Similarly, Zhang et al. introduced the Spatial Pyramid Pooling Network (SPPNet), which fuses multi-scale features for more effective crack detection. This model achieved a 2.1% improvement in mAP@0.5 and a 4.65% faster inference speed, making it well-suited for complex environments [331]. The structure of the SPPNet model is shown in Fig. 17. Zhao et al. enhanced the Path Aggregation Network (PANet) by integrating it with a crack quantification algorithm, significantly boosting segmentation performance and detection accuracy [332].

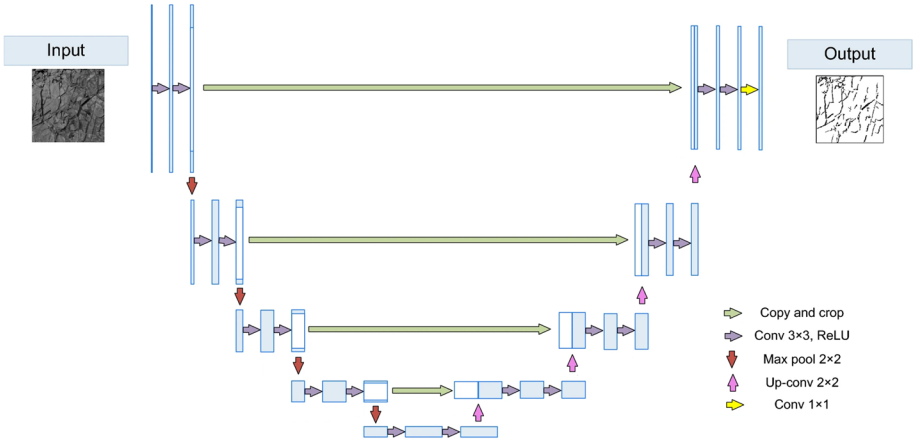


Fig. 16 Schematic of U-Net network structure [329]

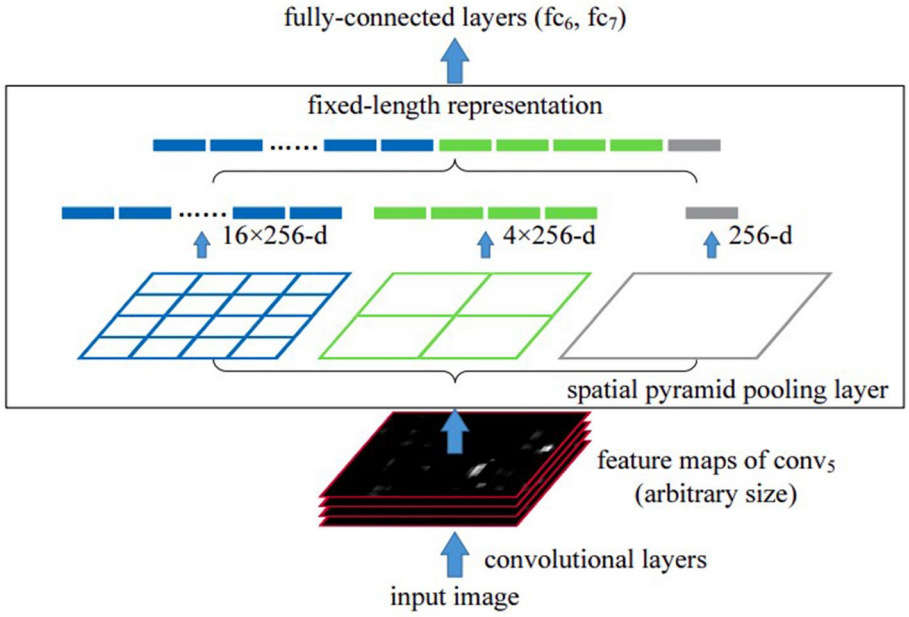


Fig. 17 Model diagram of SPPNet network [333]

Zhou et al. introduced a novel tunnel lining crack detection method using Mosaic data augmentation and MobileNetV3 for faster and more accurate single-stage networks. They further improved the model with a GAN and a channel attention module, significantly boosting performance through enhanced image data augmentation [334]. Other studies have incorporated Transformer networks' self-attention mechanisms with contextual information to address the limitations of traditional CNNs. This approach, utilizing dilated convolution and attention modules, enhances the network's sensitivity and accuracy in detecting defects of various sizes and shapes. Additionally, global average pooling strengthens the network's understanding of global features, improving defect detection in complex tunnel environments [335–337].

The choice of loss function plays a critical role in crack detection accuracy, particularly when using DL algorithms. While the cross-entropy loss function is commonly applied in conventional classification tasks, alternative loss functions are necessary for imbalanced classification problems, such as detecting blurred edges or small defects. The Focal Loss Function (FLF) [338] and Centre Loss Function (CLF) [339] have been designed to improve model performance in these challenging scenarios. Formulations for these loss functions are provided in Eqs. 6 and 7.

$$\mathcal{L}_f\left(p_t\right) = -\alpha_t\left(1-p_t\right)^{\gamma}\log\left(p_t\right) \tag{6}$$

In the Eq. 6 : $\mathcal{L}_{\boldsymbol{f}}\left(\boldsymbol{p_t}\right)$ represents the focal loss function; $\boldsymbol{p_t}$ denotes the predicted probability when the sample's true label is positive; $\boldsymbol{\alpha_t}$ is the weighting factor, typically ranging from 0 to 0.5; $\boldsymbol{\gamma}$ is the focusing parameter, which, together withp_t, forms the modulating factor$(1-\boldsymbol{p_t})^{\gamma}$.

$$\mathcal{L}_c = \frac{\lambda}{2}\sum_{i=1}^{m}\left(x_i - cy_i\right)^2 \tag{7}$$

In the Eq. 7 : $\mathcal{L}_c$ represents the center loss function; x_i is the feature of the i-th sample before the fully connected layer; c_{y_i} denotes the feature center of the class to which the i-th sample belongs; **m** is the number of samples in a mini-batch; **λ** is the balancing parameter used to control the influence of this loss function.

The Focal Loss Function addresses class imbalance by introducing a modulating factor that reduces the weight of easily classified samples, allowing the model to focus more on hard-to-classify samples, thereby enhancing its ability to distinguish between varying difficulty levels. In contrast, the Centre Loss Function minimizes the distance between feature vectors of samples within the same class, improving the neural network's discriminative power and enabling better class separation in feature representations [340].

3.4.1.2 Identification of water seepage defect features Current water seepage detection methods utilize technologies like infrared, fiber optic sensing, and high-definition imaging to identify and assess seepage extent [341]. Cheng et al. used a Fully Convolutional Network (FCN) with Visual Geometry Group-19 (VGG-19) to detect seepage intensity in tunnel images from mobile ground-penetrating radar [342]. Huang et al. applied Mask R-CNN to point cloud data from a mobile laser scanning system for automatic seepage area identification [343]. Advancements in AI, 5G, and big data have further enhanced detection capabilities. Zhao et al. used FCNs for accurate detection of defect locations in six categories of tunnel seepage images, improving segmentation accuracy by integrating a seepage detection model [344]. Fang et al. compared U-Net and PSP-Net for subway tunnel seepage detection, finding U-Net better for fast detection and PSP-Net more precise for fine-grained analysis [345]. Geng et al. proposed the Atrous Channel Pyramid Attention Network (ACPA-Net) integrated with U-Net, capturing multi-scale features for robust seepage detection [346].

Additionally, some studies have explored geometric modelling-based algorithms for seepage area recognition. Xue et al. used the structure-from-motion principle to project 2D texture-recognized defect images onto a 3D model as shown in Fig. 18, visualizing seepage defects [347]. Gao et al. introduced an FCN-RCNN model, integrating Faster RCNN, Adaptive Border ROI, and a three-layer FCN structure. This model reduced false detection rates from 0.371 to 0.0502, improving to 0.0190 after cylindrical projection correction, thereby enhancing water leakage detection accuracy [348].

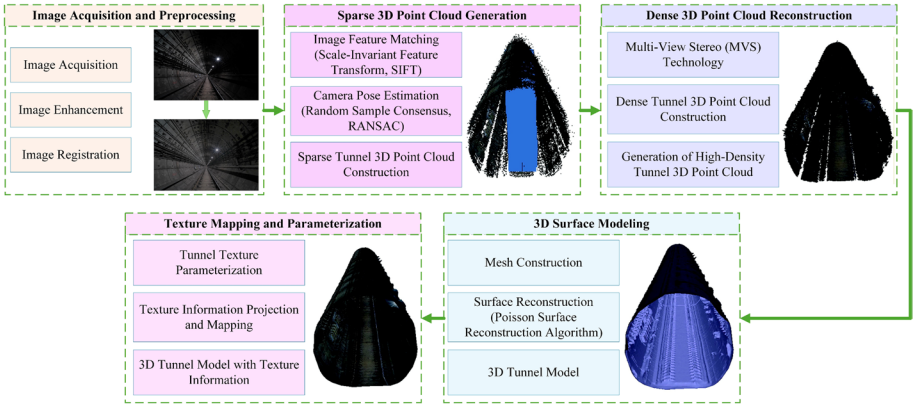


Fig. 18 Flowchart of 3D modelling of tunnel scene [347]

3.4.2 Rock burst risk prediction based on micro-seismic monitoring signals

Rock bursts, a significant geological hazard in deeply buried tunnel and caverns, pose catastrophic threats to infrastructure and personnel due to their randomness and suddenness [349, 350]. Early warning, control, and monitoring of rock burst severity are critical challenges in major construction projects worldwide, particularly in China [351]. Micro-seismic monitoring, a non-contact, three-dimensional technique, captures seismic signals from rock micro-fracturing, enabling stress state analysis of surrounding rock and has been widely used for rock burst prediction [352–354]. DL algorithms are increasingly applied to micro-seismic data to enhance the accuracy of source localization and risk prediction [355–357]. Feng et al. (2019) optimized a Probabilistic Neural Network (PNN) for predicting rock burst intensity based on micro-seismic data [358], while Wilkins et al. used CNNs for automated rock burst prediction in underground mining [359].

Di et al. explored DL algorithms on micro-seismic activity, acoustic emission (AE), and electromagnetic radiation (EMR) data (MS-AE-EMR) for rock burst forecasting, using a LSTM network to identify precursor signals and derive a risk coefficient for early warning [333, 360–362]. Tang et al. developed a DL-based system for rock burst early warning, achieving 90.1% accuracy with micro-seismic events treated as high-dimensional point clouds. However, this high accuracy must be contextualized by considering false positive and false negative rates, which are crucial for assessing the reliability of early warning systems [363]. The system's performance is particularly notable when compared to traditional methods, which typically have lower accuracy and struggle with real-time detection in complex geological conditions. In contrast, Tang et al. The SSA-CNN-MoLSTM-Attention model improves rockburst prediction by enhancing sensitivity, specificity, reducing false alarms, and achieving 93.62% accuracy outperforming traditional methods in error reduction [364]. Zhao et al. identified core technologies for micro-seismic monitoring and rock burst prediction, including multi-class classification and seismic source localization, applied in the Tianshan-Shengli Tunnel for real-time early warning [365]. Zhang et al. proposed a denoising and separation method for micro-seismic signals using a fully convolutional encoder-decoder network, enhancing rock burst prediction by filtering noise and forecasting multiple parameters for risk assessment [366, 367]. The micro-seismic multi-parameter forecasting model is illustrated in Fig. 19.

DL has demonstrated tremendous application potential in rockburst risk prediction, particularly in areas such as large-scale micro-seismic data processing, seismic source localization, and early warning of rock burst incubation [369].

With ongoing advancements in DL, sensors, human–machine collaboration, and unmanned monitoring technologies, future rockburst monitoring and early warning systems will integrate diverse data sources, including acoustic emission, stress, temperature, and electromagnetic radiation. These systems will be better equipped to handle complex geological conditions, such as high geothermal tunnel, weak surrounding rock, deeply buried tunnel under high stress, tunnel in permafrost, and Karst tunnel. These innovations will enhance risk management, improve the safety of tunnel and underground engineering and advance project management into the intelligent era.

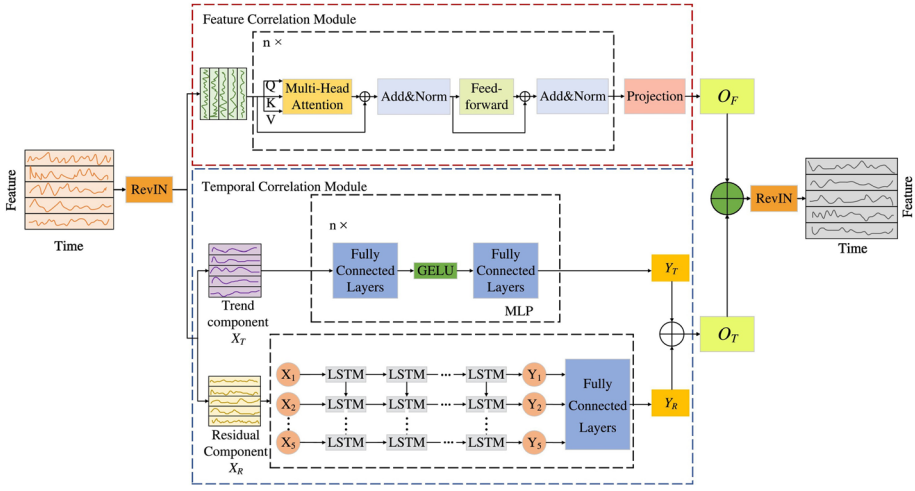


Fig. 19 Multi-parameter time series prediction model of micro-seismic [368]

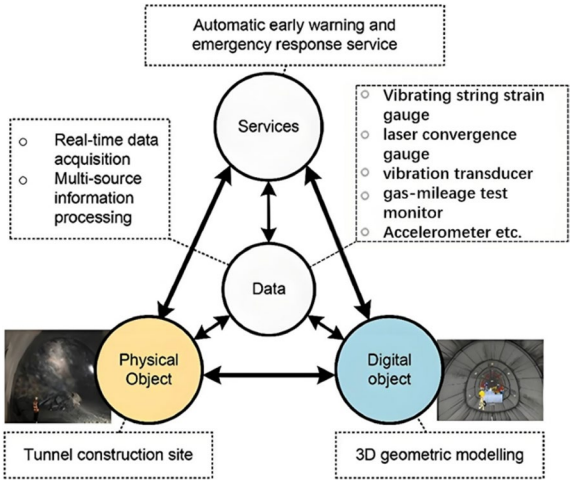


Fig. 20 Framework for intelligent disaster prevention and resilience in tunnel engineering [58]

3.4.3 Intelligent disaster prevention and real-time early warning based on multisource data

As tunnel construction projects increase and the service life of existing tunnel extends, uncertainties regarding the operational status of systems like ventilation, fire protection, and surveillance have risen. Additionally, the rapid growth in traffic volume has intensified safety concerns. Recently, resilience theory has become central in disaster risk reduction, particularly for transportation systems like road tunnel [370, 371]. This theory has gained prominence in tunnel engineering, especially in urban metro tunnel with high population mobility and mountain tunnel with dense vehicle traffic. The focus on resilience highlights the importance of ensuring worker safety during emergencies and extending the service life of construction machinery. Given the unique environmental conditions of tunnel, intelligent disaster prevention systems with real-time early warning capabilities are essential for enhancing tunnel infrastructure resilience [372]. Figure 20 shows a data-driven integration of physical and digital tunnel systems for real-time monitoring, intelligent analysis, and automated emergency response.

Tunnel and underground infrastructure disaster detection primarily relies on surveillance systems, with incident confirmation often depending on manual interpretation,

introducing subjectivity. Additionally, the lack of seamless data interoperability among subsystems such as video surveillance, ventilation control, and fire protection hinders monitoring intelligence and delays anomaly detection and emergency response [373]. With the rise of big data, DL techniques like video structuring analysis [374], object tracking [375], and facial recognition [376] have become key research directions in video processing. Anomaly detection methods based on image and information entropy enable efficient keyframe extraction, supporting intelligent disaster prevention and real-time warning systems, including dynamic fire scene reconstruction [377, 378]. Additionally, real-time detection of urban pipeline defects and tracking people and vehicles are emerging trends in smarter tunnel safety systems [379].

Key tunnel data, such as temperature, traffic volume, and CO concentration, exhibit time-series characteristics, which are crucial for predictive analytics in disaster prevention. Mienye et al. developed a smart tunnel disaster prevention decision support system that integrates multi-source data acquisition, disaster rescue decision-making, and DL data fusion based on RNNs [380]. Wu et al. used LSTM with transposed CNNs to simulate fire scenarios, predicting fire location and temperature at the tunnel crown [381]. Hong et al. created a high-resolution tunnel fire simulation database, building a deep DL model for bidirectional predictions between temperature fields and fire parameters like fire source power and ventilation speed [382]. Mao et al. proposed a robust fire detection framework using both static (flame color and texture) and dynamic features (shape and frame difference) through a multi-channel CNN [383]. For methane concentration prediction in coal seam tunnel construction, Li et al. applied an LSTM-based method, offering crucial insights for gas hazard prevention and reducing gas-related risks [384].

The appearance quality of tunnel facilities and detection of environmental anomalies are crucial for tunnel safety. Ma et al. proposed a defect detection method for underground pipelines integrating Inception and ResNet DL architectures, effectively identifying defects like misaligned joints, leakage, and corrosion, while minimizing false positives and negatives [385]. Sun et al. used DL to detect and track pantographs, catenary systems, and positioning devices from high-speed railway images, enabling real-time monitoring and early warnings to enhance operational safety [386]. Xiong et al. developed an intelligent inspection system combining visual detection and sensor data analysis for evaluating railway rails [387], using U-Net models and point cloud data processing for defect segmentation and geometry analysis [388].

Cui et al. applied DL models like PSPNet, SegFormer, and DeepLabv3 + for segment misalignment detection in shield tunnel linings, using bolt holes as reference markers to identify misalignments [389]. Zhang et al. built a heavy-haul railway rail surface image dataset, enhancing the ATSS detection algorithm with fusion pooling and texture enhancement to improve sensitivity to rail surface textures and detect obstacles near high-speed tunnel [390]. Li et al. and Wang et al. focused on supervised and unsupervised learning for the automatic identification of alarm zones and unknown intrusions in railway video surveillance, advancing railway safety technologies [391, 392].

Based on the analysis, it is clear that modifications to neural network architectures and enhanced feature extraction and integration from images and video keyframes enable DL models to effectively identify and segment various defect types across different lighting conditions and construction environments. This compensates for the limitations of traditional ML-based image detection algorithms. However, on a national scale, the

use of DL in tunnel and underground engineering is still largely experimental, with no widespread deployment of intelligent systems. As more tunnel projects enter operation, datasets will expand to include normal tunnel scenes, lining cracks, seepage, concrete spalling, and steel frame detachment. Meanwhile, innovations in intelligent detection technologies, such as 3D laser scanning, ground-penetrating radar, infrared detection, and fibre optic sensing, are enhancing integration with DL, DT technology, cloud computing, and BIM. The development of smart disaster prevention systems will support the entire lifecycle of tunnel operations, enabling resilient, intelligent maintenance and visualized management.

3.5 Tunnel structural health prediction

3.5.1 Prediction of structural deformation and stability

During tunnel excavation, energy transfer to the surrounding rock causes stress redistribution, cracks, and deformation [393]. Analyzing deformation data helps establish models linking these factors to rock mass stability, crucial for predicting tunnel deformation and ensuring safe construction. With the growth of tunnel engineering, methods for analyzing deformation and stability, such as rock mechanics and numerical simulations, have advanced [394, 395]. However, the lack of a unified framework for data acquisition and analysis often limits the sample size in data mining [396–399]. In contrast to traditional methods like the Peck empirical formula, DL offers a more effective solution to address the randomness and complexity of tunnel engineering challenges.

RNNs are widely used for predicting deformation and stability. Yuxiang et al. proposed a time series model using Nonlinear Autoregressive Neural Networks (NARNN), which accounts for dynamic factors such as excavation method and hydrogeological conditions [400]. Other models, such as the spatial–temporal fusion network by Chen et al. and the SSA-LSTM model by Hu et al., have also been applied to predict ground settlement and soil deformation [401, 402]. Further, Zhang et al. proposed an ICPO-LSTM prediction model to forecast deformation in soft rock tunnel, specifically addressing vault settlement and horizontal convergence. They optimized key hyperparameters of the LSTM network-including learning rate, hidden layer nodes, and regularization coefficients-using an Improved Crested Porcupine Optimizer (ICPO). Comparative analysis with models such as standard LSTM, CPO-GRNN, and WOA-LSTM demonstrated that the ICPO-LSTM achieved superior prediction performance, with the highest R^2 (0.9997) and the lowest RMSE (0.1700) and MAPE (0.3727%). These results highlight the model's high accuracy, robustness, and practical applicability in tunnel construction scenarios where precise deformation forecasting is critical [403]. Researchers have also used extensive field data for deformation prediction in tunnel with large cross-sections and those in challenging geological environments, improving the accuracy of stability assessments during construction [404–406].

In the training of current neural network models using monitoring data, ignoring the underlying physical principles often results in poor model interpretability and limited generalizability. To address this, Raissi from Brown University proposed the Physics-Informed Neural Network (PINN), which integrates physical models with data-driven approaches [407]. This method incorporates residual terms from physical equations into the PINN loss function, ensuring the model adheres to both observed data and

governing physical laws. Through iterative optimization, PINN allows DL models to solve complex engineering problems while remaining consistent with physical laws.

The PINN model has been applied in tunnel and underground engineering to predict ground deformation and structural behavior. PINNs address the 'black box' problem by incorporating physical laws directly into the neural network's loss function [408, 409]. This integration ensures that the model remains physically consistent with known principles, such as conservation laws or elasticity equations [410]. In tunnel engineering, PINNs improve both prediction accuracy and interpretability, offering a robust solution for predicting ground deformation and structural behavior by ensuring the model aligns with physical realities, rather than relying solely on data-driven patterns [411]. Qiujing et al. and Yang et al. integrated revised physical equations governing tunnel-excavation-induced ground deformation with deep neural networks to predict surface settlement [412, 413]. Wang et al. developed a PINN that embedded physical equations into a DL framework for real-time inversion of tunnel parameters, geological conditions, and load distribution patterns [59]. These applications highlight the practical use of PINNs in complex engineering scenarios. Furthermore, to enhance adaptability, Chang et al. combined physics-informed surrogate models with transformer-based big data models through transfer learning, as shown in Fig. 21 [414]. The integration of physical laws with DL architecture is a growing focus in future research.

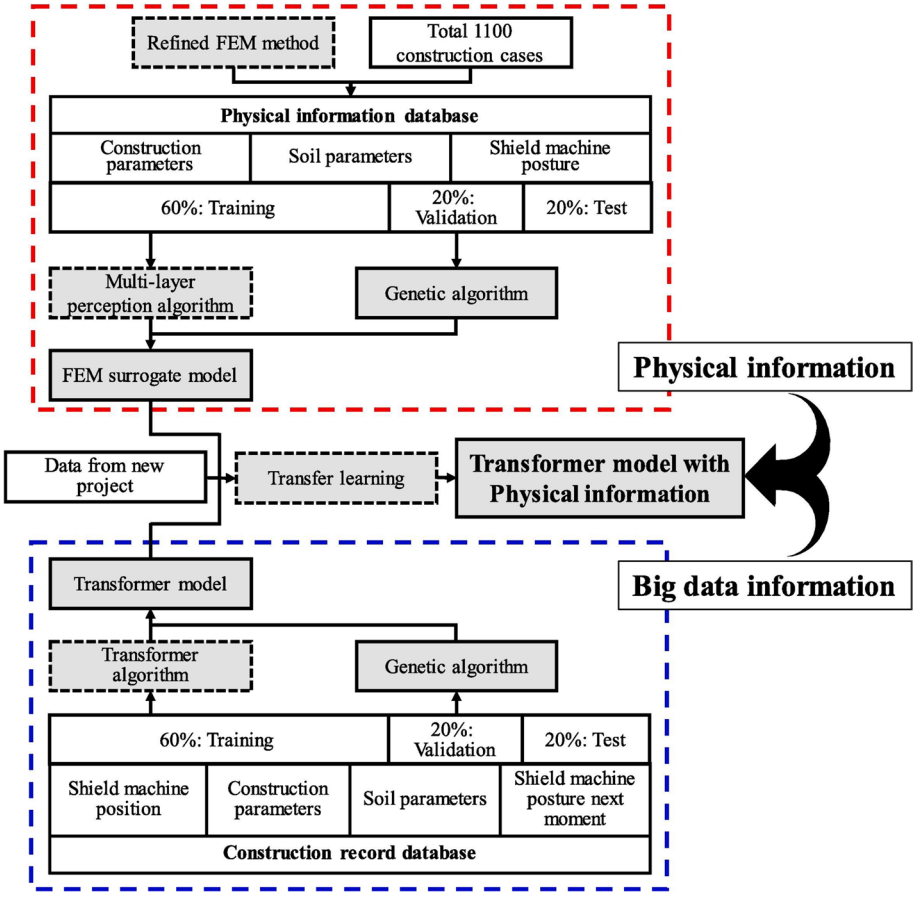


Fig. 21 Relationship framework between big data information and physical model [414]

Compared with conventional purely data-driven DL models, PINNs offer enhanced generalization and physical consistency, particularly under data-scarce or heterogeneous geological conditions [408, 409]. While traditional RNN- or LSTM-based models may achieve high predictive accuracy within single-project datasets, their performance often deteriorates when extrapolated beyond the training distribution due to the absence of embedded mechanical constraints. In contrast, PINNs incorporate governing equations and boundary conditions directly into the learning process, enabling more stable predictions under varying tunnel geometries, loading conditions, and rock mass properties [412, 413]. However, PINNs typically require careful formulation of physical residual terms and may increase computational complexity during training [407]. Therefore, rather than replacing conventional DL models, physics-guided approaches complement them by enhancing interpretability, robustness, and extrapolation capability in stability-critical tunnel applications.

Beyond deformation forecasting and PINN-based modelling, recent high-impact studies have significantly advanced machine-learning-assisted tunnel stability analysis by integrating data-driven surrogate models with adaptive finite element limit analysis (AFELA) and conventional FEM frameworks. These works explicitly address stability prediction under varying tunnel geometries and heterogeneous rock conditions, coupled tunnel–shallow foundation interaction under eccentric or inclined loading, hybrid AFELA–ML frameworks for rapid collapse assessment, and comparative evaluation of failure mechanisms using data-driven surrogates [415–417]. By training surrogate models on numerically generated limit-analysis datasets, researchers have efficiently approximated failure envelopes, plastic collapse patterns, and load-transfer mechanisms that traditionally require computationally intensive simulations. For example, AFELA–ML integrations for shallow foundations above horseshoe tunnels and square tunnel stability in rock slopes demonstrate strong predictive agreement with limit-analysis benchmarks while enabling rapid parametric exploration [416, 417]. Similarly, studies on unlined circular tunnels, surcharge loading, and complex loading interactions further highlight the capability of ML-assisted frameworks to generalize across geometric and loading variability [417, 418]. Kumar and Chauhan further reinforce this direction by examining twin tunnel stability and support requirements through limit-analysis-informed modelling, underscoring the convergence between rigorous geotechnical mechanics and data-driven predictive strategies in underground engineering safety [419].

While these contributions expand DL applications into stability-critical design domains, systematic cross-study benchmarking remains constrained by differences in failure definitions, boundary conditions, geological assumptions, and performance metrics. This heterogeneity limits direct quantitative comparison, indicating the need for standardized multi-scenario validation datasets to rigorously assess robustness and generalization in ML-assisted stability modelling.

3.5.2 Prediction of mechanical behavior in underwater tunnel

Underwater tunnel face higher construction risks, greater environmental uncertainty, and more complex geological challenges compared to land-based tunnel, making geotechnical exploration and monitoring more difficult and resulting in lower accuracy in obtaining mechanical parameters [420, 421]. The limited number of completed underwater tunnel worldwide also restricts the training of DL models for deformation

prediction. To address this, researchers have focused on integrating DL algorithms with Structural Health Monitoring (SHM) techniques, which are primarily used for predicting mechanical behavior and evaluating tunnel stability [422]. However, SHM has developed more slowly than big data technologies and is not yet widely applied in practice. Traditional time-series models often fail to capture essential patterns in the large, complex, and diverse SHM data, leading to poor predictive performance. The introduction of DL algorithms into SHM allows for more accurate predictions of underwater tunnel behavior under multiple influencing factors.

Tan et al., using the Wuhan Yangtze River Tunnel as a case study, applied the Autoencoder algorithm and RNN to analyse the spatial correlations and temporal dependencies of structural evolution, using segment strain, tunnel temperature, and water levels as input parameters. They proposed a DL model called the Autoencoder Fused Temporal-Spatial-Load Network (ATSLN), which effectively predicts future trends in structural behavior by integrating external load data and spatiotemporal information [60]. Their results revealed that short-term structural responses, sensor locations, internal tunnel temperature, and external water pressure are key factors influencing structural evolution. Additionally, they explored GCNs as shown in Fig. 22, which can extract features from neurons and neighbouring nodes, establishing temporal dependencies between data points [423]. Tan et al. further enhanced this approach by combining GNNs with the GRU algorithm in a multi-task learning model to predict strain behavior and crack propagation in underwater shield tunnel segments [60].

DT technology integrates surface hydrological and subsurface construction environments by linking 3D tunnel and geological models. This enables dynamic simulation of the construction cycle in a virtual space, reducing the impact of natural variability on tunnel stability and mechanical behavior. The main challenge, however, is efficiently retrieving and applying the required engineering knowledge for specific performance predictions.

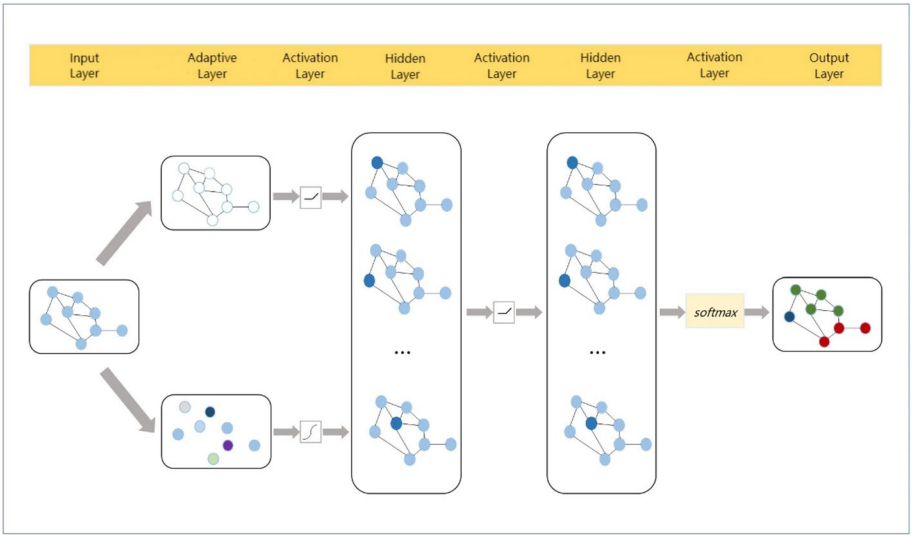


Fig. 22 Structure of graph convolutional network [424]

3.6 Surrounding rock classification

3.6.1 Classification methods based on rock mass rating indicators

Classifying the surrounding rock is crucial for tunnel design and construction. Analyzing the exposed rock mass at the tunnel face helps assess excavation safety, determine protective measures, and guide shotcrete and steel support placement [425, 426]. It also enables the prediction of geological conditions ahead by examining rock layers, fractures, and groundwater, which informs subsequent construction operations [427, 428].

At the end of the eighteenth century, Russian scholar Werner proposed a qualitative rock classification method based on hardness. The modern development of rock mass classification began with Terzaghi's rock load classification system, introduced in 1946 [429]. Over the past 200 years, many studies have contributed to various classification methods. To standardize rock mass stability evaluation, Chinese experts adopted the Basic Quality (BQ) value and established the GB/T 50218-2014: Classification Standard for Engineering Rock Mass. While most classification methods combine quantitative and qualitative approaches, in practice, surrounding rock classification often relies on subjective judgment, leading to misclassification, especially when field personnel lack adequate expertise [430]. As tunnel and underground cavern construction advances to greater depths, current surrounding rock classification methods face challenges such as low data utilization, poor accuracy, and limited decision-making intelligence under complex geological conditions. A recent study by Kilic et al. addressed these challenges by applying a clustering-guided Light Gradient Boosting Machine (LightGBM) algorithm to classify lithology in soft ground tunnel. Their approach was particularly effective in handling imbalanced and heterogeneous geological datasets during TBM excavation, offering a scalable and interpretable solution for lithological classification in complex conditions [431].

DL algorithms can extract features from tunnel face images. At the same time, cloud computing can process large-scale data, prompting the development of a geological information management platform for surrounding rock classification [426]. Wang et al. proposed a qualitative classification framework for expressway tunnel using a BP neural network enhanced with adaptive learning and momentum algorithms to classify surrounding rock during design and construction [432]. Ma et al. further optimized the BQ-based model by integrating joint orientation, dip angle, and rockburst intensity, developing an intelligent classification system for the Nibashan Tunnel that combines qualitative and quantitative assessments [433]. The workflow and application of this method are illustrated in Fig. 23.

3.6.2 Classification methods based on rock mass feature images

With the increasing diversification of geological monitoring data and enhanced GPU capabilities, researchers have begun using core and tunnel face feature images for qualitative and quantitative surrounding rock classification [434, 435]. Liu Feiyue et al. utilized Mask R-CNN to preprocess core drilling images, enabling automatic RQD calculation for more refined rock mass characterization, thus expanding DL's application in intelligent classification [436]. Several studies processed on-site images to capture rock features such as fractures, water inrush, and crushed zones [437, 438], using models like AlexNet [439] and ResNet for classification and rock mass identification [440]. MySQL databases were integrated into tunnel engineering classification systems, achieving good

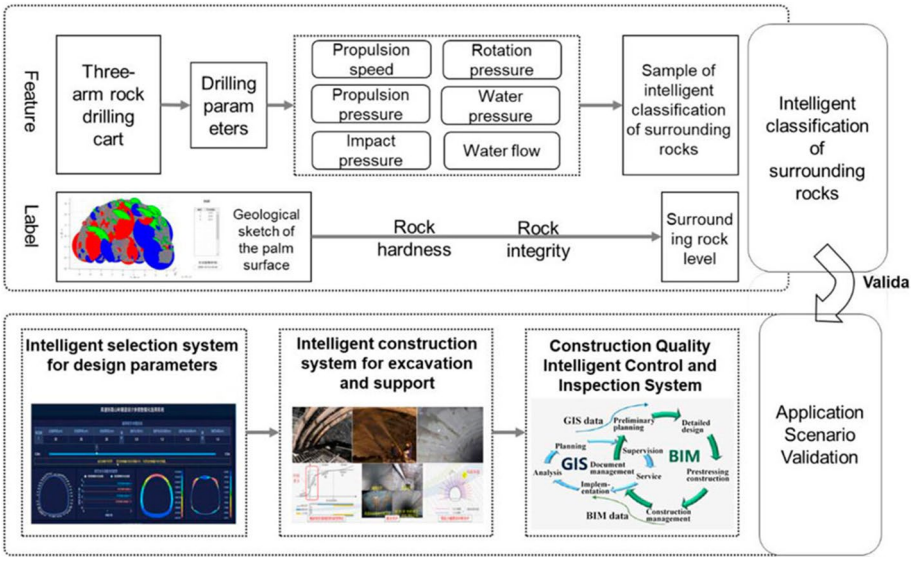


Fig. 23 Flowchart of intelligent surrounding rock classification method and its application scenarios [61]

identification performance [32]. Cheng et al. used drilling data and penetration rates as input for a Bidirectional LSTM model, optimising performance with stratified k-fold cross-validation, thus advancing rock classification in tunnel engineering [62].

In tunnel surrounding rock classification, using high-definition digital cameras and advanced geological forecasting techniques to analyse tunnel face geometric features is crucial for intelligent classification [441, 442]. Smith and Paraskevopoulou employed 3D point cloud technology to create detailed tunnel face models and used the U-Net neural network for automatic joint trace recognition, comparing 2D U-Net, 3D KPCONV, and XGBOOST methods, with U-Net performing best, especially with limited data. Their study integrated the BQ classification method for assessing rock quality, offering an automated solution for tunnel condition evaluation [443]. They proposed a method combining multi-source heterogeneous data, including 1D time-series data, 2D rock structure images, and 3D geophysical data, as shown in Fig. 24. Additionally, excavation machine thrust depth and cutterhead torque were used as feedback factors for rock-mechanical perception, employing a self-organizing neural network to predict and classify surrounding rock levels [444, 445]. With large volumes of surrounding rock data, time- CNNs can process the data and analyse long-term memory, significantly improving the prediction of quantitative analysis indicators for rock classification [446].

Rock mass classification is vital in rock engineering, aiding in the assessment of rock strength and stability. Research shows significant progress in applying DL to surrounding rock classification. CNNs are used to extract spatial features, such as joint orientation and color, from rock mass images. RNNs model the sequential nature of these images, facilitating time–space analysis of rock mass data. By integrating remote sensing technology, additional rock mass information from satellite images improves the accuracy and reliability of classification.

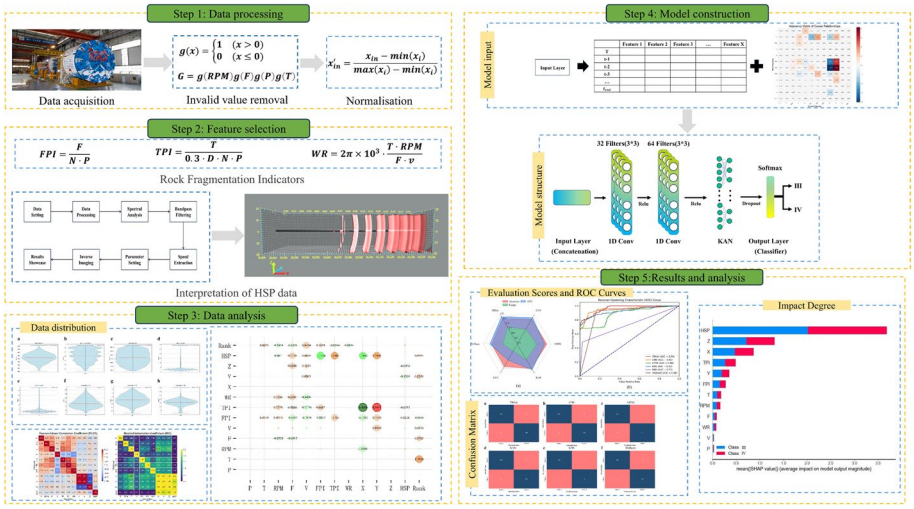


Fig. 24 Intelligent classification method of surrounding rock based on multi-source heterogeneous data fusion [447]

3.7 Tunnel face image recognition and classification

3.7.1 Joint fissure identification

The tunnel face provides the clearest reflection of engineering geological conditions [448], and quickly acquiring image data from tunnel faces is crucial for ensuring safe tunnel construction [323]. However, construction environments, camera limitations, and workers' photography skills often result in blurry images, uneven lighting, and incomplete data [449, 450]. Traditional ML techniques struggle with issues such as low joint continuity and poor clarity, which hinder efficient, intelligent construction. The advent of DL algorithms has shifted the focus to image recognition for analyzing tunnel face rock mass structures, addressing challenges such as automatic quality assessment and joint identification [451, 452]. Mandal et al. created a tunnel face image dataset, classifying images under varying conditions and using models like DenseNet169, ResNet50, and VGG16 to assess image quality based on clarity and similarity [453].

The research indicates that optimizing tunnel face image quality involves improving DL network structures, annotation methods, and expanding dataset distribution. Li et al. used U-Net and CliqueNet CNNs to extract and recognize contour features and fine cracks on tunnel faces, refining crack length and inclination for automatic feature extraction [63]. Chen et al. proposed a method for automatic segmentation and quantification of rock mass fissures, utilizing the FraSegNet pixel segmentation model, chain code polyline trajectory traversal algorithm, and polyline approximation skeleton extraction algorithm [454]. The FraSegNet model incorporates a dilated Spatial Pyramid Pooling (ASPP) module to detect multi-scale features using varying dilation convolution rates [455], as shown in Fig. 25. The chain code algorithm identifies key feature points in crack images, simplifying their representation with chain codes and performing approximation for multiple segments. In contrast, the polyline extraction algorithm outputs key coordinates for each crack's pixel, capturing trace features [456].

Peng et al. incorporated the PANet into Mask R-CNN to address low detection accuracy in tunnel face images under dark tunnel conditions, enabling rapid joint position detection [457]. Lei et al. combined the Transformer attention mechanism and deformable convolutional networks with Mask R-CNN, enhancing feature extraction by

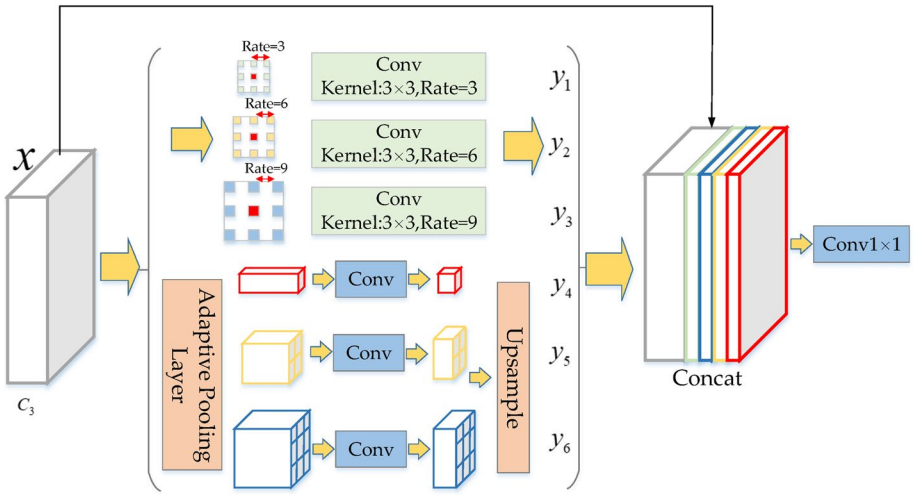


Fig. 25 Atrous spatial pyramid pooling model [456]

focusing on key areas and improving the network's adaptability to geometric deformations through deformable convolution kernels. This combination significantly boosted Mask R-CNN's performance in recognizing and quantifying joint and crack features on tunnel faces [458].

3.7.2 Rock mass lithology classification

Scientific studies have also explored automatic classification of rock mass lithology using DL. Recent studies, the Inception-ResNet-V2 model is one of the commonly used DL algorithms. This model combines the efficient feature extraction capability of the Inception network with the deep residual learning ability of ResNet, allowing it to effectively adapt to different image resolutions and varying environmental conditions when processing rock mass images [440]. Zhao et al., based on transfer learning techniques, used Inception-ResNet-V2 as the training model, to classify geological masses in cross-section images. It achieved an impressive overall accuracy of 99.72% in identifying rock mass lithology, including granite, limestone, basalt, and shale [64]. Qin et al. collected 42,100 tunnel face images under various conditions, including different temperatures, humidity levels, brightness, and dust concentrations, and used Inception-ResNet-V2 to train the model [459]. They successfully identified the rock structure present in the dataset, achieving automated classification of tunnel face rock structures.

Gupta et al. employed the lightweight MobileNet V2 model to efficiently identify the lithology of rocks, such as gneiss, granite, and sandstone, with a focus on reducing computational load and storage resource consumption while ensuring robust performance [460]. Fu and Wang investigated the use of Mask R-CNN for mineral identification, specifically for recognizing minerals like feldspar, quartz, and biotite in rock thin sections. The model achieves 97.6% accuracy in ore identification, demonstrating its effectiveness in both classification and localization within challenging mining environments [461]. Chen et al. integrated dilated convolutions and depth-wise separable convolutions into a DL model, proposing the DeepLabv3 + model based on deep CNNs. They established a database containing 32,040 images of limestone, dolomite, loess, and red clay to train, validate, and test the model's performance, which was applied to the recognition and detection of weak interlayers in rock tunnel faces [462].

Based on the analysis above, it is evident that transfer learning techniques and the ImageNet dataset are indispensable elements in the task of rock lithology recognition and classification in tunnel face images. For improving the accuracy of image recognition and classification, there are generally three methods: (1) Selective emphasis on key features in the image to minimize interference from secondary information; (2) Compensation of image pixels to make the improved image as close as possible to the original image; (3) Optimizing the network structure, where models that combine shallow detailed features and deep semantic features extraction can output images with higher resolution and stronger informational features.

Across the seven identified domains, a clear methodological and functional evolution is evident in the application of DL within tunnel and underground engineering. Early research primarily focused on feature extraction and pattern recognition using CNNs and RNNs to process single-source data, such as tunnel imagery or sensor readings. With advancements in computational capacity and data acquisition, subsequent studies have shifted toward hybrid and multi-modal architectures that integrate visual, acoustic, and numerical simulation data for more comprehensive prediction and classification tasks. These emerging models demonstrate the capability to learn cross-domain correlations and to perform real-time adaptive control under complex geological and operational conditions. Conceptually, the seven domains are not isolated research silos but represent an interconnected continuum—from data-driven perception and monitoring to physics-informed, autonomous decision-making systems. This integrative synthesis provides coherence across thematic divisions and situates each technological trajectory within a unified developmental framework for intelligent and resilient tunnel engineering.

4 Existing problems

Although various DL algorithms can address similar problems, differences in network structures, limitations of principles, and data diversity lead to inconsistent outcomes. The following issues remain:

(1) Optimizing neural network structures involves multiple factors. For fully connected networks, increasing hidden layers alone does not enhance accuracy and may even cause overfitting. Additionally, although CNNs exhibit excellent performance in image processing and recognition, obtaining high-resolution and high-quality images of lining structures in tunnel engineering is often costly. Challenges faced by RNNs include information loss and limitations in processing relatively simple data types.

(2) Data acquisition and sharing remain major limitations. DL in tunnel engineering requires large, diverse datasets, with tabular models needing thousands of rows and vision/time-series models requiring tens to hundreds of thousands of samples [463, 464]. Challenges like limited, noisy, and imbalanced data are addressed through methods like K-means-guided SMOTE, LightGBM, and quantile transformation, along with class-weighted losses and threshold tuning [431]. In practice, obtaining sufficient training data requires multi-source sensing (e.g., GPR, CCTV, LiDAR) and cross-project data pooling-yet such systems are costly and rarely standardized. Moreover, high-performance computing infrastructure (e.g., GPUs, cloud services) is necessary to train complex models, and field deployment often demands real-time inference with lightweight, optimized networks. The lack of unified platforms for data

sharing and real-time processing hinders dynamic decision-making, especially in small to mid-scale projects.

(3) Although DL algorithms for tunnel lining defect detection and tunnel face image recognition have matured, their application in risk assessment and performance prediction remains limited by the "black box" nature of these models [20, 465]. This lack of interpretability reduces the transparency of model predictions, making it difficult to build trust in critical safety applications. Explainable AI methods, such as SHAP and LIME, are emerging as essential tools to address this issue by providing more understandable and interpretable model outputs, which can improve decision-making processes [466]. The inability to explain model predictions could hinder the adoption of DL in safety–critical areas like tunnel construction, where regulatory requirements may soon mandate transparency and interpretability for algorithms involved in risk assessment, safety monitoring, and operational predictions. Therefore, increasing the interpretability of DL models is crucial to meet industry standards and regulatory demands, enhancing their practical applicability and fostering broader acceptance in engineering practice.

(4) Despite the significant potential of DL algorithms in tunnel and underground engineering, their real-world deployment faces challenges, particularly in complex geological conditions [20]. These methods require integration with existing monitoring systems, which are often based on traditional techniques, and the generalization ability of DL models is limited in new, unseen geological contexts-such as deep-buried tunnel with high ground stress and varied rock and groundwater conditions [467–469]. This diversity in geological features can reduce prediction accuracy and reliability, especially when models have not been validated in similar environments. Long-term case studies and performance data are essential to assess the robustness of these algorithms in real-world scenarios, and future research should prioritize improving model generalization, integration with real-time systems, and validation across diverse geological conditions to enhance their practical application in engineering decision-making [230].

4.1 Fundamental research challenges

Despite growing adoption, DL in tunnel engineering faces key barriers that limit scalability, reliability, and field deployment. The most pressing challenge is the lack of diverse, high-quality datasets. Over two-thirds of the 494 reviewed studies relied on synthetic or project-specific data (Table 2), restricting model generalization across varying geological and construction conditions. This directly impacts transferability-many models degrade when applied outside their training context, yet few adopt domain adaptation or transfer learning to address this.

Beyond transferability concerns, robustness under varying tunnel geometries, loading eccentricities, slope conditions, and heterogeneous rock mass properties remains insufficiently examined in current research. In most reviewed studies, model performance is validated within single-project datasets or controlled numerical settings, with limited cross-geometry or cross-geology testing. As a result, robustness is often inferred indirectly from stable performance under restricted conditions rather than systematically evaluated across diverse engineering scenarios. The absence of standardized multi-configuration benchmark datasets and unified robustness metrics further constrains

objective assessment of DL reliability under realistic underground variability, representing a fundamental methodological challenge for future research.

Interpretability is another critical gap. In risk-sensitive applications like deformation or failure prediction, most models offered limited transparency beyond visualizations, lacking causal explanations essential for operational trust. Purely data-driven models also dominate, with minimal integration of physical laws. Though PINNs offer promise, hybrid approaches remain rare, undermining extrapolation in complex or evolving ground conditions.

Deployment viability is constrained by computational demands. High-performing models often require significant resources, making them impractical for real-time use on-site. While compression and edge optimization have been explored [57, 58], scalable deployment pipelines are largely absent. Multi-sensor fusion also remains underdeveloped, with most studies relying on single-modality data and lacking robust integration frameworks for uncertain or noisy environments.

Finally, regulatory readiness is lacking. Few studies engage with certification, ethical safeguards, or BIM integration, and most fail to propose feedback mechanisms for DTs or real-time control. Addressing these gaps requires a coordinated agenda that prioritizes data sharing, interpretability, physics-informed modelling, and governance structures to unlock the full potential of AI in tunnel systems.

4.2 Critical evaluation and limitations

Figure 3 shows a steady upward trend in publications since the introduction of DL in 2006, with a notable surge after 2019. This increase can be attributed to growing recognition of deep learning's potential, advancements in AI and computational tools, more international collaborations, and increased research funding for large-scale infrastructure projects. Each publication was classified according to the Scimago Journal Rank (Q1–Q4 and unranked) to reflect the distribution of research quality. Publication output rose from 9 papers in 2016 to a peak of 105 in 2024, with 99 publications recorded up to November 2025, indicating that the upward trend remains ongoing. The marked post-2019 surge also reflects the widespread adoption of GPU-accelerated computing, the availability of open-source AI frameworks, and the proliferation of sensor-driven tunnel monitoring and digital-twin systems that enabled large-scale data generation. The increasing share of Q1–Q2 publications since 2021 demonstrates a transition from exploratory studies to rigorously validated, application-oriented research, signalling the growing maturity of DL applications in tunnel and underground engineering.

Firstly, many studies report strong predictive accuracy but lack rigorous validation protocols. For example, the ES-Caps-FCN model achieved a remarkably low MSE of 0.059866% and R^2 of 99.813% in parameter inversion tasks [51], yet its training relied entirely on simulated datasets with no field validation, limiting its practical applicability. Similarly, the BP–FLAC3D model yielded high accuracy ($R^2 = 0.991$) on synthetic samples, with stress prediction errors under 5% [52], but again lacks generalization testing under real geological variability. This over-reliance on synthetic or homogeneous data introduces serious risks of overfitting, especially in tunnelling contexts where lithological, hydrological, and stress conditions vary dramatically.

Secondly, the issue of generalization remains insufficiently addressed. A significant proportion of models are trained on small or single-project datasets, such as

the 1D-CNN used for predicting cutter wear on Japanese soft-ground tunnel, which achieved $R^2 = 89.6\%$ and MAE = 1.6 [54]. However, its applicability to different geological formations or tunnel methods remains unverified. In another case, the LSTM–GWO model used only 1,125 samples and showed $R^2 = 0.9795$ and MAPE = 0.009% for predicting tunnel advance rate, but its dataset was constrained to a specific project with uniform lithology [53]. These instances underscore that high accuracy on localized datasets does not equate to robust generalization.

Moreover, comparative evaluations across studies often lack fairness due to dataset and context heterogeneity. The reviewed literature uses diverse input parameters, output variables, and evaluation metricsranging from MAE and RMSE to F1-score and mAP@0.5making it difficult to assess algorithmic superiority objectively. For instance, in defect detection, YOLOv5 and YOLOv8 models achieved mAP@0.5 values of 86.5% and 95.3% respectively [57, 58], but were trained on different image resolutions, radar sources, and defect types, reducing the fairness of direct comparisons.

Reproducibility is another major concern. Across the reviewed articles, very few shared their datasets or complete model configurations. Even in high-performing studies, such as those integrating CNNs with Bayesian inference for safety assessment (81.3% accuracy, 100% recall) [55], there is often an absence of code repositories, data annotations, or standardized benchmarking protocols. This severely hampers validation, replication, and broader scientific engagement.

In terms of methodological transparency, some studies employ complex black-box models (e.g., GANs, GNNs) without adequately addressing model explainability or interpretabilitycrucial aspects for deployment in high-risk engineering environments. For example, defect detection studies using EfficientNet integrated with YOLOv8 achieve strong results (Precision = 94.7%, Recall = 93.9%) [58], yet offer minimal insight into decision rationale, which could be critical for infrastructure risk communication and operational control.

The most pressing issue, however, is the inadequate treatment of geological complexity and model robustness in variable conditions. Tunnel engineering is uniquely vulnerable to subsurface unpredictability, but only a minority of studies attempt to simulate or validate under diverse scenarios. For example, even top-performing models like PIOELM for structural health monitoring (reducing MSE by 50.57–85.29%) are constrained by assumptions of linear elasticity and do not incorporate plastic or time-dependent deformation [59]. Moreover, quantitative assessments of how geological variability impacts model error or failure rates remain largely absent in the literature.

Finally, several studies fail to contextualize ML challenges within the tunnel domain. Issues like data imbalance, signal noise, or temporal drift are described generically, without mapping them to specific tunnelling construction phases such as face advancement, segment erection, or grouting operations. While the manuscript identifies challenges like data scarcity and poor generalization [20], it often does so descriptively rather than analytically. A systematic root-cause analysis-linking, for instance, poor model robustness to over-reliance on synthetic datasets or inadequate sampling of heterogeneous lithologiesis missing.

In summary, despite technical advancements, DL applications in tunnel engineering are constrained by insufficient methodological scrutiny, limited real-world validation, and a lack of domain-specific criticality. To ensure robust, transferable, and safe

deployment, future research must adopt standardized evaluation protocols, promote open datasets, incorporate uncertainty quantification, and ground algorithmic design in the spatiotemporal and physical realities of tunnelling construction systems.

5 Conclusion

This review synthesizes a decade of advancements (2016–2025) in applying DL across seven interconnected domains of tunnel and underground engineering: (1) parameter inversion, (2) construction machinery optimization, (3) construction and operation control with risk assessment, (4) tunnel safety monitoring and defect detection, (5) tunnel structural health prediction, (6) surrounding rock classification, and (7) tunnel face image recognition and classification. Drawing from 494 peer-reviewed studies, the findings reveal a pronounced shift toward intelligent, data-driven tunnel systems, with DL models demonstrating substantial improvements in predictive accuracy, automation efficiency, and real-time monitoring capabilities. However, although reported performance metrics are frequently impressive, the predominance of simulation-based validation underscores the need for broader field-level benchmarking and long-term engineering verification to ensure practical robustness and cross-context reliability.

In parameter inversion, models such as ES-Caps-FCN and BP–FLAC3D hybrids achieved exceptional accuracy ($R^2 > 0.99$) and drastically reduced computation timeup to 200 × faster than traditional simulations [51, 52]. Yet, these models were often trained on synthetic datasets, limiting their real-world reliability and geological generalizability. In construction machinery optimization, time-series models like LSTM–GWO and 1D-CNN demonstrated R^2 values of 0.97–0.98 [53, 54], indicating strong predictive capacity for operational parameters like advance rate and torque. However, most applications were tunnel-specific and lacked robustness across mixed ground conditions and machine types.

In the domain of construction and operational risk assessment, hybrid models integrating CNNs, Bayesian inference, and DT platforms showed improved real-time decision-making potential, with precision and recall metrics exceeding 90% in some cases [55, 56]. Still, they often suffered from single-case validation and limited multisource data fusion. The domain of defect detection emerged as one of the most mature, with YOLOv8 achieving mAP@0.5 = 95.3% and model compression strategies (e.g., YOLOv5 with distillation) reducing size by over 80% while improving accuracy [57, 58]. Yet, issues such as lighting variability, occlusion, and small-target detection in real-world environments persist.

In structural health prediction, models like PIOELM reduced MSE by up to 85.3%, and GC-GRU achieved RMSE = 15.52 με and a 45-day forecast horizon [59, 60]. However, these models often omitted environmental or dynamic loading conditions, which are essential for tunnel longevity assessments. Surrounding rock classification, using XGBoost and Bi-LSTM with SMOTE or ADASYN balancing, showed classification accuracies of up to 90%, but was limited by small, imbalanced datasets from single tunnel projects [61, 62]. Finally, in tunnel face image recognition, U-CliqueNet and Inception-ResNet-V2 achieved mean IoUs of 86.96% and accuracy of 99.72%, respectively [63, 64], demonstrating DL's power in visual geological mapping, though real-time deployment remains constrained by image quality and computational overhead.

Beyond empirical findings, this review proposes a novel algorithm-to-process mapping framework that links DL architectures to tunnel-specific engineering tasks based on data modality and process dynamics. For example, CNNsused in over 60% of defect detection and image classification studiesare well-suited to spatial pattern recognition; Bi-LSTM and GRU networks capture excavation dynamics and sequential processes; GANs improve model robustness in low-data conditions; GNNs model relational dependencies across sensor networks and spatial tunnel systems. This mapping provides a mechanistic foundation for architecture selection and contributes a theoretical lens specific to underground infrastructure, addressing critiques of abstraction in existing AI reviews.

Despite these advances, several systemic barriers persist: (1) limited use of real-field data and absence of shared, annotated benchmark datasets; (2) poor generalization across tunnel types, machine systems, and geological conditions; (3) minimal use of interpretable models in safety–critical environments; and (4) weak coupling between DL models and real-time operational control. These challenges are not just technical-they are deeply infrastructural and data-governance-related.

To bridge current limitations in tunnel engineering AI, future research should focus on federated learning for secure multi-project collaboration, PINNs for embedding physical laws into models, and GNNs for capturing spatial sensor relationships. Causal inference methods are needed to distinguish true risk factors from correlations, while integrating deep models into DTs enables adaptive, real-time control. Finally, advancing explainable AI-using SHAP or attention-based toolswill be essential for trustworthy deployment in safety critical environments. Together, these directions form a practical and scalable path toward intelligent, resilient, and interpretable tunnel infrastructure systems.

Abbreviation

ML	Machine Learning
DL	Deep Learning
AI	Artificial Intelligence
AFELA	Adaptive finite element limit analysis
MLP	Multilayer perceptron
CNN	Convolutional neural network
GNNs	Graph neural networks
GANs	Generative adversarial networks
RL	Reinforcement Learning
DRL	Deep Reinforcement Learning
DT	Digital Twin
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
GPR	Ground Penetrating Radar
CCTV	Closed-Circuit Television
ES-Caps-FCN	Enhanced Spatial Capsule-based Fully Convolutional Network
DNN	Deep Neural Network
LR	Linear Regression
MSE	Mean Squared Error
MAE	Mean Absolute Error
R^2	Coefficient of Determination
MRE	Mean Relative Error
RE	Relative Error
SVR	Support Vector Regression
GWO	Grey Wolf Optimizer
ANN	Artificial Neural Network
PSO	Particle Swarm Optimization
UQ	Uncertainty Quantification
BNNs	Bayesian Neural Networks
SMOTE	Synthetic Minority Oversampling
FCN	Fully Convolutional Network
BPNN	Back-Propagation Neural Network

UCS	Uniaxial Compressive Strength UCS
RMR	Rock Mass Rating
RQD	Rock Quality Designation
Bi-LSTM	Bi-directional Long Short-Term Memory
TBM	Unnel Boring Machine
NATM	New Austrian Tunnelling Method
SVM	Support Vector Machine
NDT	Non-Destructive Testing
SPPNet	Spatial Pyramid Pooling Network
RMSE	Root Mean Squared Error
PINN	Physics-Informed Neural Network
SHM	Structural Health Monitoring
GCN	Graph Convolutional Network
VGG	Visual Geometry Group
ResNet	Residual Network
BQ	Basic Quality
LightGBM	Light Gradient Boosting Machine
ADASYN	Adaptive Synthetic Sampling

Author contributions
C.Z, M.H, J.W, A.A, M.H.S: conceptualization, writing-review, data collection, manuscript drafting and supervision of the study; C.Z, J.W: investigation, validation; A.A, M.A, B.M, Y.Z: design, data collection; G.M.S, M.F: investigation, editing; M.A, M.S.R: data analysis, editing; C.Z: editing, supervision.

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Data Availability
No datasets were generated or analysed during the current study.

Declarations

Competing interests
The authors declare no competing interests.

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